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Methane fluxes in a thawing permafrost peatland in Finnmark, Norway

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Lena Kaiser

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Submitted to:	Faculty of Biology, Chemistry and Earth Science, Chair of Soil Ecology
Supervisor:	Prof. Dr. Werner Borken
Second Assessor:	Dr. Cristian Estop Aragonés, University of Münster, Institute of Landscape Ecology, Ecohydrology and Biogeochemistry Research Group
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Declaration of Authorship

I, Lena Kaiser, hereby declare that I have written this present thesis

“Methane fluxes in a thawing permafrost peatland in Finnmark, Norway”

on my own and have not used any sources or aids other than those stated in the references. All parts taken verbatim or in substance from published or unpublished sources are identified as such. This thesis has not been submitted to any other examination authority in the same or a similar form.



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Abstract

Methane fluxes in permafrost peatlands play an important role in global greenhouse gas budgets since they can act as net sink for atmospheric methane. However, due to global warming, permafrost peatlands are increasingly exposed to thawing, which transforms them from methane sinks into sources, thereby amplifying the rise in atmospheric methane concentration. The objective of this thesis was to investigate methane fluxes in two different palsa structures (intact and degraded) and a recently collapsed thermokarst pond in a thawing permafrost peatland in Finnmark, northern Norway, to gain a better understanding of methane dynamics in this changing ecosystem. Flux measurements were conducted using the closed dynamic chamber method and a portable greenhouse gas analyzer. Intact (- 2.35 mg CH_4 m^{-2} d^{-1}) and degraded (-2.09 mg CH_4 m^{-2} d^{-1}) palsas acted as a net methane sink, while the water saturated thermokarst was a net source of methane (80.7 mg CH_4 m^{-2} d^{-1}). During the short period between mid-July and beginning of September, there were only minor fluctuations in methane flux rates. Emissions from thermokarst were highest mid-season and dropped rapidly mid-August. Soil temperature, volumetric water content (VWC) and thaw depth were not correlated with methane fluxes from palsas. Thermokarst emissions were influenced by VWC and soil temperature. Further, in an incubation experiment, aiming to measure CH_4 fluxes in different peat layers at same temperature, methane oxidation rates were highest in the upper 20 cm with VWC as the main driver of CH_4 fluxes. These results suggest that degraded (deeply thawed) palsas continue to serve as effective sinks for atmospheric methane; however, further subsidence of the peat body due to ongoing thawing processes and its transformation into thermokarst will turn this sink into a source of methane.

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1 Permafrost peatlands in the subarctic region

Subarctic and boreal peatlands cover an estimated area of 3.46 x 10^6 km², representing 87 % of the world's peatlands, while many of them contain at least sporadic permafrost (Van Huissteden, 2020; Vitt, 2006). Based on spatial continuity, sporadic permafrost is defined by 10-50% frozen subsoil in a region, occupying 17 % of the total permafrost region (Van Huissteden, 2020). In contrast, continuous permafrost occurs in areas where 90 % or more of the land surface is underlain by permafrost. Permafrost is defined as ground, either soil or rock, that remains at or below 0 °C for 2 consecutive years (Van Huissteden, 2020). The active layer at the soil surface thaws each summer and refreezes in winter, typically ranging from a few decimeters to over a meter in thickness; its base defines the permafrost table (Van Huissteden, 2020). Peatlands are ecosystems where organic production by arctic plant communities, mainly mosses facilitating peat formation (Turetsky et al., 2012), exceeds decomposition, forming peat layers of at least 40 cm. Despite limitations in carbon accumulation due to low temperatures, nutrient scarcity (especially nitrogen) and short, light-limited growing seasons, such systems can develop into permafrost peatlands when underlain by perennial frozen ground (Van Huissteden, 2020). Permafrost peatlands in the northern circumpolar permafrost zone are suggested to store 277 Pg carbon, excluding the underlying mineral soil C (Schuur et al., 2008). Typical morphological structures in the sporadic permafrost zone are palsas; they occur in many areas of the subarctic, e.g. northwestern and southwestern Canada, in European Russia and in western Siberia, when mean annual ground temperature is around 0 °C and limited precipitation with < 800 mm/year (Borge et al., 2017; Martin et al., 2019). They are characterized as elevated, dry peat mounds with a frozen peat core causing frost heave (Van Huissteden, 2020). Their spatial extent varies between 1 and 7 m in height and less than 100 m in diameter (Harris et al., 1988). Larger frost heave features in peat bogs, expanding over hundreds of meters, are described as peat plateaus (Van Huissteden, 2020). Palsas need specific environmental conditions for their formation, namely long-lasting air temperature below 0 °C, thin snow cover, and low summer precipitation. Physical properties of the peat are responsible for their development and persistence since thermal conductivity in dry peat is low and insulates the ice core during hot and dry summers, resulting only in minor thawing. On the contrary, summers characterized by humid and wet weather lead to increased thawing and may impair the frozen ice core due to high thermal conductivity of wet and frozen peat (Seppälä, 2011). As soon as precipitation begins in autumn, peat becomes saturated and freezes, resulting in higher thermal conductivity, enabling deep infiltration of winter frost. In winter, a thinner snow cover due to wind abrasion or other local conditions allows frost to penetrate deeper. The elevated surface has an amplifying effect on maintaining the ice below since peat above the initial frozen core remains drier and affects the insulating capability. Equally, a thinner snow cover forms in winter on top of the mound, enabling deeper freezing. These soils containing ice are preserved if the heat balance avoids the ice from thawing, while mechanical

instability and genesis of cracks, enabling water and heat forwarding further inside the palsa lead to its collapse (Van Huissteden, 2020). Disturbance and palsa degradation is also attributable to global warming; e.g. the area covered by palsas and peat plateaus has decreased by up to 70 percent since the 1950s in Finnmark, northern Norway, and is now estimated to 110 km² (Borge et al., 2017). Permafrost thaw is characterized as abrupt (decadal scale) or gradual (decadal to centennial scale) (Grosse et al., 2011). The latter is known as active layer deepening, resulting in increased soil moisture and enhanced anaerobic conditions, while also vegetation, carbon cycle and peat chemistry change (Hodgkins et al., 2014; Turetsky et al., 2020). Landforms resulting from abrupt soil subsidence triggered by thawing of ground ice are often referred to as thermokarst lake or thermokarst pond, displaying waterlogged conditions including vegetation and peat chemistry change as well. However, they do not have much in common with the typical “karst” features, resulting from dissolution of limestone in non-permafrost environment (Van Huissteden, 2020). Current model projections of permafrost environments remain highly uncertain due to insufficient representation of key physical processes. However, continued degradation at present-day rates would likely lead to the widespread disappearance of palsas and peat plateaus in northern Norway in the 21st century (Borge et al., 2017). This degradation is closely linked to ongoing climatic changes, as rising air temperatures, increased winter precipitation, and longer growing seasons are expected to further enhance ecosystem CH_4 emissions (Parmentier et al., 2015; 2013).

In this context, methane emissions from permafrost systems originate from three main sources: recently produced organic matter as part of ecosystem respiration, thawed ancient carbon from Holocene or Pleistocene deposits beneath thaw lakes and ponds, and deeper intra- or sub-permafrost gas and hydrate reservoirs, though the latter are not typical for permafrost peatlands (Van Huissteden, 2020). Current estimates suggest that methane emissions from permafrost regions contribute only a small amount (in the magnitude of ~ 0.5 %) to the total CH_4 budget but are expected to increase gradually, while releases from gas hydrates are likely negligible (0-1 %) (Dean et al., 2018). Methane emissions driven by anaerobic decomposition of recently produced organic matter are suggested to dominate over methane emissions from previously frozen carbon (Cooper et al., 2017).

Moreover, the release of methane following permafrost thaw represents a critical positive climate feedback, as methane has a 28–34 times higher 100-year global warming potential (GWP) than carbon dioxide (Knoblauch et al., 2018). However, considerable uncertainty remains regarding the overall magnitude of CH_4 emissions, as hydrological and landscape changes associated with permafrost thaw and their effects on CH_4 production and uptake are difficult to predict (Voigt et al., 2019).

1.1 Methanogenesis

As thawing of permafrost continues and intact palsas are transitioning into degraded palsas and eventually into thermokarst with higher water saturation and less oxygen supply, the system may shift from a methane sink to a methane source, where CH_4 is released into the atmosphere (Kjær et al., 2024). The anaerobic production of methane (methanogenesis) depends on several factors, such as the oxidation state of the organic carbon, the availability of alternative electron acceptors (nitrate, ferric iron, sulfate, organic matter) due to microbial competition (Gao et al., 2019; Knoblauch et al., 2018) and soil moisture (chapter 1.3) as well as temperature (chapter 1.4) (Lu et al., 2022). Methanogenesis only occurs under anaerobic conditions and the absence of other reducible inorganic electron acceptors. When O_2 is no longer available, the other electron-accepting compounds are utilized first as these reactions provide higher energy yields for microorganisms. After depletion of inorganic oxidants, methanogenesis as the least favorable electron-accepting reaction can proceed (Conrad, 2020). In a drying and rewetting cycle, with reoxidation of reduced iron and sulfur, electron acceptor depletion and methanogenesis might occur to a certain extent coextensively, likely due to micro-environments that enable individual respiration pathways, but enhanced methanogenesis only proceeds after depletion of electron acceptors (Knorr & Blodau, 2009). In addition to that, there are two main pathways of methanogenesis performed by microbes that are classified in the Archaea domain of life. Acetoclastic methanogenesis is the conversion of the methyl group of acetate to CH_4 (equation 1), also known as acetate fermentation. The other pathway is the reduction of CO_2 with electrons derived from hydrogen (H_2), known as hydrogenotrophic methanogenesis, as displayed in equation 2. Hydrogen gas acts as the main electron donor for CO_2-reducing methanogens, while other CO_2-reducing microbes from the bacteria domain also use H_2 to produce acetate that subsequently serves as substrate for acetoclastic methanogens (Ferry, 2010).

$$CH_3COO^- + H^+ \rightarrow CH_4 + CO_2 \quad (1)$$

$$CO_2 + 4\,H_2 \rightarrow CH_4 + 2\,H_2O \quad (2)$$

Once methane is produced in an anaerobic environment, it is making its way up through the soil column to the surface. Four methane transportation pathways can be distinguished: Firstly, gas diffusion describes the slow movement of gas molecules through the pores in the soil; methane is either dissolved in the soil solution or flows via air-filled pore spaces. This process represents the normal transport movement in the soil and is dependent on soil gas concentration gradients in the soil. Secondly, the convective flow is driven by pressure gradients between soil air and the atmosphere, contributing only a minor flux to the total emissions. These gradients are developed by wind eddies breezing over the soil surface, or differences in temperature, alterations in

barometric pressure and adaptations in pore water content to evapotranspiration, infiltrating rain or soil freezing. Thirdly, ebullition refers to the transportation of gas bubbles through water-filled soil pores or the water column until the soil surface or water table is reached (Van Huissteden, 2020). Emission rates caused by ebullition are comparable to those created by the diffusive flux, yet they might induce episodically high rates in water-logged soils (Baird et al., 2004; Turetsky et al., 2008). Genesis of CH_4 bubbles is subject to the poor solubility of methane in water, which is an order of magnitude lower compared to CO_2 ($3.3 * 10^{-4}$ mol m^{-3} Pa^{-1} for CO_2; $1.4 * 10^{-5}$ mol m^{-3} Pa^{-1} for CH_4 at standard atmospheric pressure) (Sander, 2015; Van Huissteden, 2020). Besides, as temperatures increase, gases become less soluble, turning it into a climate-related indicator in warming soils (Van Huissteden, 2020). Lastly, plant-mediated transport involves the movement of gases through aerenchymatous tissues in the roots and stems of wetland plants. It is suggested that gas transport by aerenchymal structures makes up between 30 and 100 % of the methane flux (Bhullar et al., 2013), however these structures also enable oxygen supply to the roots, which is then utilized by aerobic microorganisms including methanotrophs which can oxidize significant amounts of methane during the transport to the atmosphere (Walter & Heimann, 2000). Once transported to near-surface zones, whether via diffusion, convective flow, ebullition, or plant-mediated pathways, a portion of CH_4 is oxidized by methane-oxidizing bacteria abundant in the aerated upper soil layer (Van Huissteden, 2020).

Lower temperatures in permafrost soils are one main difference compared to temperate wetland soils, however the abundance and composition of methanogenic community is similar (Wagner & Liebner, 2009). A distinct vertical zonation is predominant in permafrost soil columns, with more psychrophilic archaea at depth in the soil, near the permafrost table. Activity of psychrophilic methanogenic archaea has been discovered at temperatures below zero (Rivkina et al., 2000). Such findings underline that methane production can occur even under harsh environmental conditions, thereby contributing to the overall methane balance in cold ecosystems.

1.2 Methane oxidation

Methane oxidation is considered the only biological sink to atmospheric methane. Methane release from permafrost is moderated by methanotrophs, which oxidize an estimated percentage of 20–60% of this methane before released to the atmosphere (Singleton et al., 2018). Aerobic methanotrophs belong to the group of gram-negative bacteria that use methane as carbon and energy source (Trotsenko & Murrell, 2008). Methane oxidation proceeds as displayed in the equation below (Trotsenko & Murrell, 2008):

$$CH_4 + 2\,O_2 \rightarrow\ CO_2 +\ 2\,H_2O \tag{3}$$

Environmental conditions such as soil moisture affect the uptake of methane since oxygen availability depends on the moisture content which is in turn altered with permafrost thaw (Kolb, 2011; Perryman et al., 2020). Additionally, fluxes of CH_4 that take place near the soil surface are rate limited by the diffusion gradient between the concentrations of CH_4 in deeper layers and the concentration in the ambient atmosphere. The diffusion of soil gases within the soil profile is controlled by the volume and structure of the air-filled pores within the soil column which depends on the occurrence of water in the pore space of the soil, which will fluctuate depending on soil structure and soil texture. The substrate availability (i.e., CH_4 and O_2 concentrations at depth) for methanotrophic bacteria might thereby be either directly or indirectly affected by these abiotic characteristics (Juncher Jørgensen et al., 2024).

Two groups of methane oxidizing bacteria are differentiable, namely type I and type II methanotrophs. They are defined by their ability to cope with different methane concentrations, meaning that they have distinct methane affinities. In an experiment with diffusion columns, where the concentration of methane in the upper zone was low and the concentration of oxygen was high due to ambient conditions, a prevalence of type I methanotrophs in zones near the top of the columns was indicated. On the contrary, the dissolved oxygen concentrations were very low at the bottom of the columns, and methane concentrations were high. Results indicated that these bottom zones contained type II methanotrophs. Consequently, it is suggested that type I methanotrophs outperform type II methanotrophs at low methane concentrations whereas the growth of type II methanotrophs was predominant under low-oxygen and high-methane conditions (Hanson & Hanson, 1996). This suggests that type I methanotrophs may exhibit a higher methane affinity than type II methanotrophs (Liebner & Wagner, 2007).

Type I and type II methanotrophs behave also differently regarding temperature conditions. A study by Liebner & Wagner (2007) indicate that type I methanotrophs dominated throughout the active layer profiles but their number strongly fluctuated with depth. Hence, upper soil layers are suggested to be predominantly populated with psychrotolerants or a diversified community of mesophiles and psychrotolerants. In contrast, type II methanotrophs were constantly abundant through the whole active layer and displaced type I methanotrophs close to the permafrost table and thus are assumed to be psychrophilic. On the other hand, other studies (e.g. Rothschild & Mancinelli (2001)) state that microbial communities do not prefer the cold, but rather can survive extreme conditions.

Furthermore, methanotrophs require bioavailable nitrogen (N) to preserve their metabolism and N is the main nutrient limiting plant and microbial growth in Arctic ecosystems. One main opponent for N resources, in particular ammonium (NH^{4+}), are nitrifiers causing direct competition between methanotrophs and nitrifiers for N substrates. Thus, as dissolved N concentration increases, methane uptake was shown to be higher (Voigt et al., 2023).

However, aerobic methane oxidation is not the only pathway to control methane emissions from peatland soils. Significant contributions of anaerobic oxidation of methane (AOM) to

the aerobic methane oxidizing bacteria (MOB) have been identified especially with nitrite, nitrate and Fe (III) as electron acceptors, aiming that AOM has the potential to be a control on net CH_4 fluxes from northern peatland soils (Fu et al., 2025; Miller et al., 2019). Yet the significance of this oxidation pathway to peatland CH_4 cycling globally is less clear than that of aerobic methanotrophy (Perryman et al., 2020) and AOM are less common in these environments which reflects in minor rates of methane oxidation (Singleton et al., 2018).

The net methane flux, either emission or uptake, which is released into the atmosphere or removed from it, is the result of the difference between the methane produced by methanogenesis and the amount consumed by methanotrophs, since both processes act simultaneously in the soil column where methanotrophs are active in an aerated transitional zone at the boundary with atmosphere (Hanson & Hanson, 1996). Under well aerated conditions, soils act as a net methane sink, constituting an important control on CH_4 emissions to the atmosphere (Kolb, 2011).

1.3 Influence of soil moisture on methane fluxes

The environmental conditions for CH_4 production are determined by moisture content (both in thermokarst and palsas) since prolonged water saturation creates anaerobic conditions and low redox potential after depletion of alternative electron acceptors, creating favorable conditions for methanogenic communities (Smith et al., 2018). A water level above the soil surface was found to promote the activity of methanogens, while under the same moisture treatment, the cumulative CH_4 emissions increased with the increase of temperature (Lu et al., 2022). This suggests that moisture is the main factor controlling CH_4 emissions, while temperature enables higher emissions under already oxygen-limited and thus water-saturated conditions (chapter 1.4). Song et al. (2020) also concluded that hydrological conditions following permafrost thaw control emissions, as the cumulative methane production in incubations from the active layer was 64-times higher in saturated anaerobic conditions than from natural-aerobic conditions. Moreover, in their incubation experiment they noted a delay in CH_4 production which is due to differences in soil properties, environmental conditions and microbial community composition. For instance, in the anaerobic incubation the maximum CH_4 production occurred approximately after 100 days after starting, then continued at a high level for 60 days and then declined. This corresponds to other findings, where low initial CH_4 production rates have been measured repeatedly because of lack of methanogenic microbes (Knoblauch et al., 2018; Wagner et al., 2007). However, when considering long-term experiments (> 7 years), these results suggest that there will be continuous CH_4 production when stable methane producing microbial communities are established in permafrost (Knoblauch et al., 2018). Moreover, when anaerobic conditions prevail, methane production is mainly controlled by the availability of terminal electron acceptors and microbial activities (Song et al., 2020).

Contrastingly, increased soil moisture inhibits methane oxidation by reducing gas diffusivity. This process depends on air-filled porosity and governs the movement of the gases to and from the atmosphere (Smith et al., 2018). However, if soil becomes too dry, methane oxidation decreases, as microbial activity is inhibited (Smith et al., 2018). Drought impacts microbial physiology primarily through osmotic stress, which damages biomolecules such as cell walls, membranes, enzymes, and DNA (Thomsen et al., 2025). Microbes counteract this stress by modifying structural compounds, repairing damaged molecules, or entering dormancy (Thomsen et al., 2025). Beyond osmotic effects, drought restricts microbial access to carbon and nutrients by impairing diffusion, mass transport, and cellular movement, as soil pore connectivity diminishes and thereby alters microbial activity and carbon cycling (Thomsen et al., 2025). There is rather an optimal water content at which methane oxidation reaches its maximum (Juncher Jørgensen et al., 2024). Soil moisture also influences methane oxidation in combination with a depth related gradient since depth depicts a proxy for oxygen availability and diffusion of air into the soil in the same way as this applies to soil moisture. Generally spoken, concentration profiles of CH_4 (as substrate for methane oxidation) in in situ soil columns exhibit low atmospheric concentrations at the surface with decreasing concentrations with depth within several centimeters, however in greater depths CH_4 concentrations increase again, exhibited as a product of methanogenesis (Juncher Jørgensen et al., 2024). Conversely, oxygen is abundant at the surface and limited in deeper layers (Hanson & Hanson, 1996). As a result, maximum methane oxidation rates occur at the depth offering the optimal balance between sufficient O_2 and CH_4 influx from the atmosphere and adequate CH_4 availability emitted from the active layer (Juncher Jørgensen et al., 2024). However, studies assessing methane oxidation potentials in deeper peat layers under oxic conditions remain scarce, limiting the understanding of the vertical distribution of methane oxidation, particularly under varying oxygen and moisture conditions across the peat profile.

Permafrost thaw, abrupt (decadal scale) or gradual (decadal to centennial scale) (Grosse et al., 2011), results in the formation of thermokarsts or degraded palsas, respectively; with differing hydrology and moisture conditions unknown under future climate scenarios. How conditions of permafrost thaw (abrupt vs. gradual) affects microbial community and thus CH_4 fluxes remains a major uncertainty in warming subarctic regions (Ramage et al., 2024). Thus, quantifying the amount of CH_4 produced or oxidized in permafrost soils remains necessary for understanding future methane balances in permafrost peatlands.

1.4 Influence of temperature on methane fluxes

Temperature controls methanogenesis and methane oxidation by affecting methanogenic and methanotrophic bacteria (Whalen, 2005). Whalen (2005) suggested a significant effect of temperature on CH_4 production rates due to the wide range of Q_{10} (reaction rate increase for a 10 °C temperature increase). Voigt et al. (2017) investigated the influence of warming on palsas

(vegetated and unvegetated plots, rather dry) using hexagonal open-top chambers for artificial warming-up (mean increase in air temperature of 0.95 °C) with control-plots set up next to it. They found a significant warming effect on both the bare peat and vegetated peat plateaus due to increased fluxes. Especially in the bare peat plateaus, warming resulted in an almost fivefold increase in CH_4 emissions in the first year and more than doubled in the second year. Flux data were averaging at 0.21 mg CH_4 m^{-2} d^{-1} at control plots and 0.75 mg CH_4 m^{-2} d^{-1} from warmed bare peat surfaces. In comparison, the vegetated peat plateaus turned from a small CH_4 sink to a source with warming in both years with an average flux of 0.12 mg CH_4 m^{-2} d^{-1} (Voigt et al., 2017). Moreover, the gene abundance of methanogens is supposed to be more sensitive to temperature rise than that of methane oxidizing bacteria, which suggests that the emission intensity of soil CH_4 in peatlands might be more closely related to the activity and number of methanogens (Lu et al., 2022). Similar findings were made by Treat et al. (2014), where aerobic and anaerobic treatments at 20 °C differed largely in methane emissions, with more CH_4 being released under anaerobic conditions, but no differences at -0.5 °C which can be traced back to lower temperatures. Below freezing, microbial respiration and liquid water are limited. Additionally, at – 0.5 °C a net uptake of CH_4 under aerobic conditions and a near-zero production under anaerobic conditions were recognized (Treat et al., 2014).

This weak temperature sensitivity of methanotrophs is further evidenced by low Q_{10} values (Smith et al., 2018), which are in the range of 5.3-16 for methanogenesis and 1.4-2.1 for methane oxidation (Dunfield et al., 1992). Voigt et al. (2023) showed that significant response of CH_4 oxidation to temperature was perceptible at a higher temperature range (4 °C VS 20 °C) in laboratory incubations under optimal dry conditions that neither limited O_2 diffusion nor induced drought stress. This is further underpinned by Liebner and Wagner (2007), where maximum oxidation potentials were detected at 21 °C, while the lowest temperature with maximum methane oxidation was 4 °C, performed by mesophile / psychrotolerants and psychrophiles, respectively, indicating that methane oxidation potential varies gradually across a broad temperature range rather than responding strongly to a e.g. 10 °C increase. The weak temperature response further likely reflects substrate limitation, as the scarce methane supply, resulting from diffusion barriers within soil and the low atmospheric CH_4 concentration, constrains the extent to which temperature can further stimulate oxidation activity (Smith et al., 2018). Other studies also reported no effect of temperature to methane oxidation, which they attributed to either substrate-limited conditions or that other important controls on CH_4 oxidation have a prevailing influence (Juncher Jørgensen et al., 2024; Liebner & Wagner, 2007; Voigt et al., 2023). Voigt et al. (2023) conclude that temperature becomes an important secondary control under favorable substrate and moisture conditions.

However, the temperature response of organic soils in permafrost regions is still less well constrained compared to mineral soils (Lupascu et al., 2012). It remains unclear how warming affects the balance between CH_4 production and oxidation, as temperature not only enhances

active layer deepening and microbial activity, but also alters precipitation regime and hydrologic conditions in northern regions (Dean et al., 2018). Moreover, studies explicitly addressing how progressive thaw over the course of a growing season influences seasonal methane flux dynamics are limited.

1.5 Aim of thesis and hypotheses

Given the considerable climatic impact of methane emissions and their substantial uncertainties outlined above, understanding the controls and drivers of methane fluxes under changing environmental conditions is crucial. Therefore, the aim of this thesis was to identify methane fluxes in a thawing permafrost peatland in northern Norway with their typical landforms palsas and thermokarsts. In particular, the primary objective was to understand how thaw-induced degradation affects methane fluxes and to determine the role of environmental variables in this process.

Therefore, the following is hypothesized:

1) Methane fluxes change in the transition from intact palsas to degraded palsa to recent thermokarst structures. On intact negative fluxes (uptake of atmospheric CH_4) are expected, but the sink strength for atmospheric CH_4 is lower in degraded palsas. The recent thermokarst acts as a net CH_4 source (CH_4 emission).

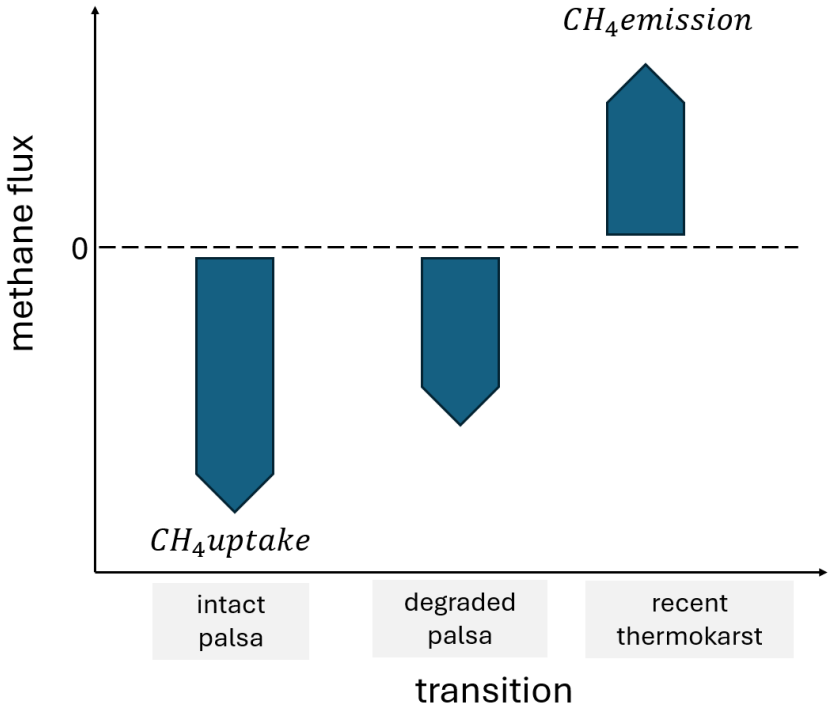


Figure 1: Schematic illustration of hypothesis 1 - methane flux rates are hypothesized to be different in intact / degraded palsas and in recent thermokarst, altering from a sink of atmospheric methane in palsas to a source of methane in the thermokarst.

2) Due to an increasing thaw depth on palsas during course of the season, methane fluxes will increase with increase in peat layer available for methanotrophic activity and thus enhanced diffusion of atmospheric CH_4 into the soil.

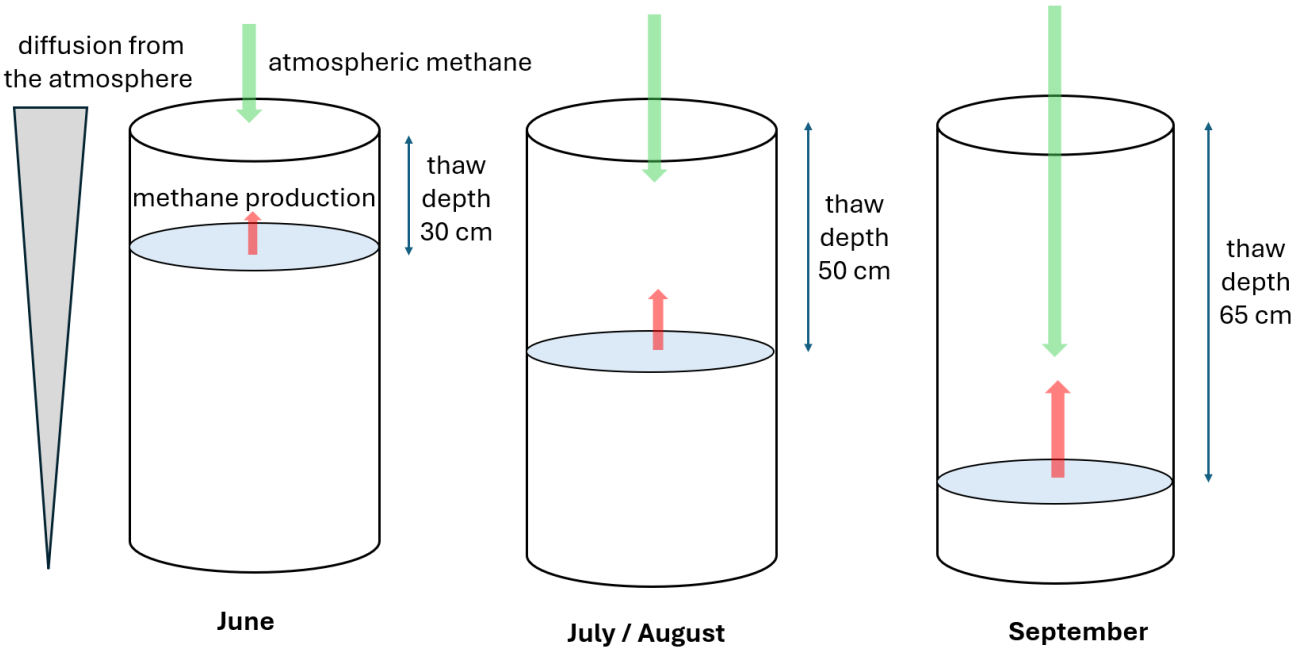


Figure 2: Schematic illustration of hypothesis 2 - as thaw depth increases throughout the season, an increasing layer of peat becomes available for methanotrophic bacteria and diffusion of atmospheric methane, thus leading to increased methane oxidation rates.

3) With an ex-situ (3 a) and an in-situ (3 b) experiment, the influence of peat depth on methane fluxes (both positive and negative) is indicated.
    a. Methane uptake decreases with increasing depth due to decreasing oxygen supply and higher moisture contents. Highest methane uptake is expected in the upper ten centimeters of palsa peat layer, whereas methane production is anticipated in the deeper layers close to the permafrost layer.

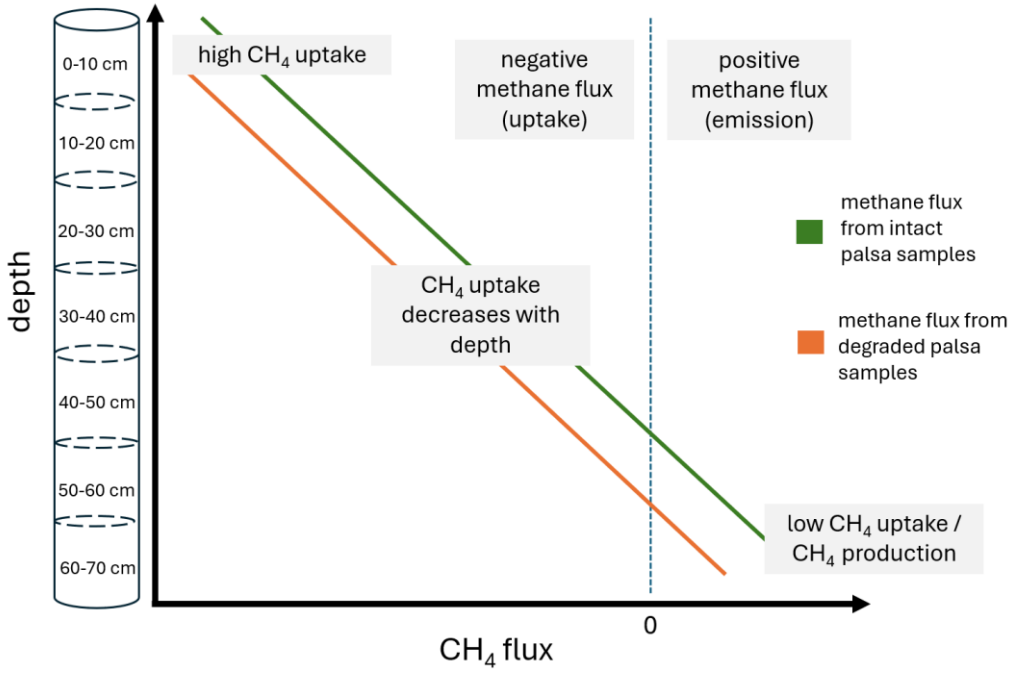


Figure 3: Schematic illustration of hypothesis 3 a - methane uptake decreases with increasing peat depth, while methane production is expected in deeper layers. Colors distinguish intact (green) and degraded (orange) palsa samples, with lower methane uptake expected in degraded samples.

b. In in-situ conditions, methane uptake is higher in sealed than in unsealed treatments, implying that methane production from deeper peat layers contributes to net fluxes in unsealed treatments.

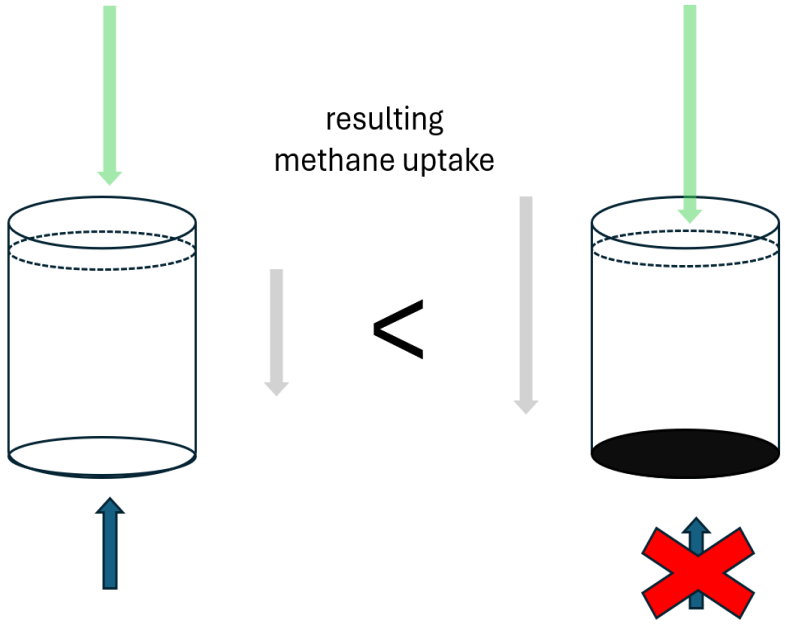


Figure 4: Schematic illustration of hypothesis 3 b – higher methane uptake is expected in sealed treatments, since methane production from deeper peat layers is excluded.

2 Materials and Methods

2.1 Site description

The field campaign was carried out from mid-June to mid-September 2025. The research site, Áidejávri (Lat = 68.74744722, Lon = 23.32675095, 400 m a. s. l.), was located in Finnmark, the northernmost county of Norway (Figure 5). The research area was deglaciated approximately 10,900 to 10,800 cal. yr BP (calibrated years before present) (Stroeven et al., 2016).

From a geological perspective, the area is underlain by Precambrian bedrock, which is overlain by glacial and glaciofluvial sediments deposited during Pleistocene glaciations. In the mid-Holocene, wet fen conditions prevailed in areas suitable for such ecosystems, replaced by a shift towards drier surface conditions in the late Holocene, associated with a gradual transition from nutrient- and mineral-rich fen conditions towards intermediate fen stages before they give way to raised dry bogs. After approximately 750 cal. yr BP, plant macrofossil analyses indicate permafrost aggradation, which coincides well with the Little Ice Age cooling after 850 cal. yr BP. This led to frost heave and thus formation of palsas or peat plateaus. As a consequence of frost heave, the upper peat layer became aerated and also enabled more extensive decay in the upper peat, resulting in lower peat accumulation rates (Hichens-Bergström & Sannel, 2023).

Due to climate change, which caused an increase in air temperature by 1.0 °C to 1.5 °C in Finnmark, the land area in northern Norway containing palsas and peat plateaus declined since the 1950s, with a total decrease in area between 33 and 71 % (Borge et al., 2017). As a result of degradation of permafrost and melting of ground ice, the genesis of thermokarst land-forms, e.g. thaw ponds or collapsed bogs, is initiated (Olefeldt et al., 2021).

Mean annual temperature (MAT) and precipitation (MAP) in Áidejávri ranged between -3.9 to -0.1 °C (mean -1.9 °C) and 294.4 to 790.9 mm (mean 475 mm) in the period from 1991 to 2020 (reference). During time of field work, mean temperature ranged from 5.5 to 21.4 °C (mean 12.8 °C) and mean precipitation from 0 to 16.7 mm (mean 1.7 mm). The data was taken from the Norwegian Meteorological Institute measured at the Šihččajávri weather station (Lat: 68.7559654º N, Lon: 23.5395347º E), 9.4 km from Áidejávri. Data for precipitation during the period of field work originates from the station in Kautokeino, 31.7 km from Áidejávri (Lat: 68.9964º N, Lon: 23.0349º E), as the closer one was not available due to technical issues.

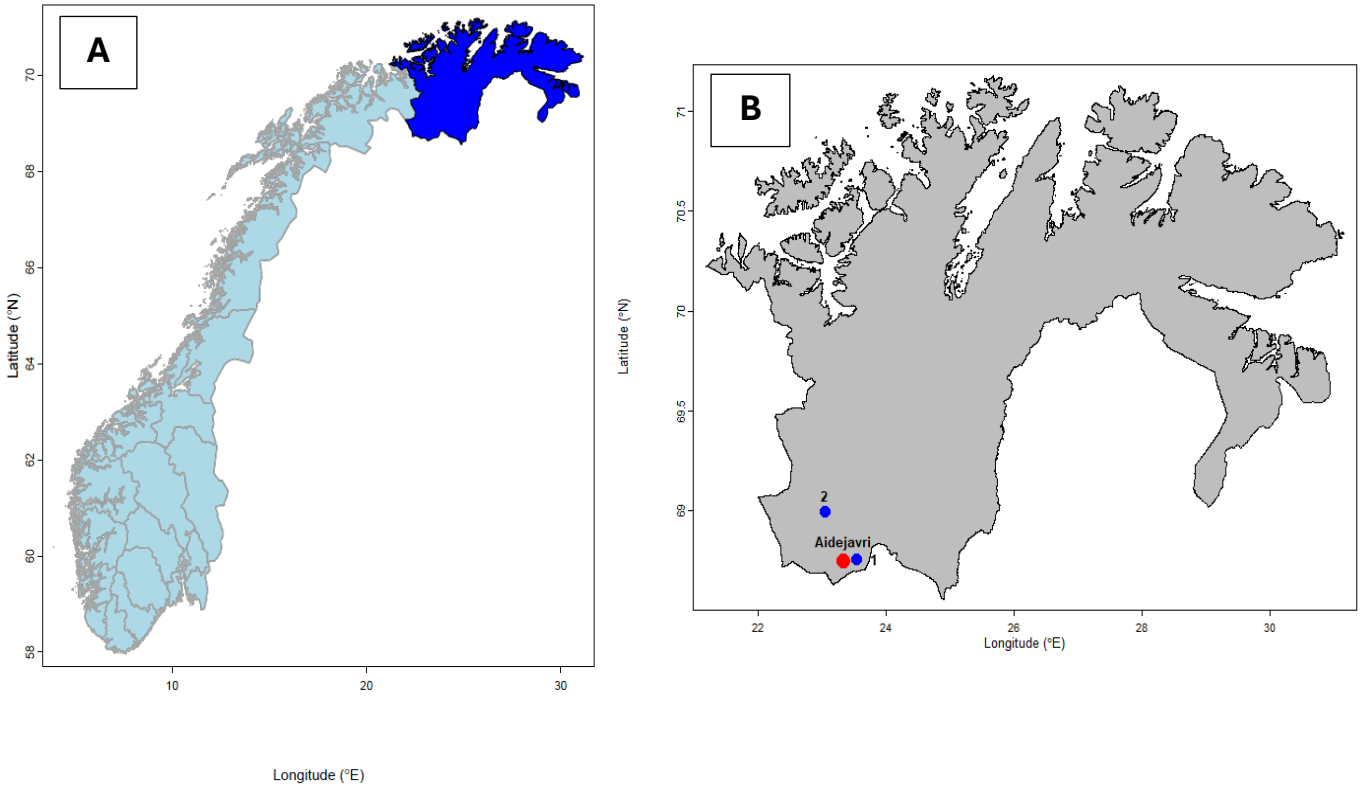


Figure 5: (A) map of Norway and the Finnmark county highlighted in darker blue (B) map of the Finnmark county where the study site lays within (map outlines taken from gadm.org). The numbers represent the climate stations. 1 is the Šihččajávri station and 2 is the Kautokeino station.

The continental subpolar research site is surrounded by several small streams and consists of many intact palsa mounds alternating with thermokarst depressions and intermediate stages between those structures: these transition stages are characterized by active layer deepening, reaching lower thaw depths than intact palsas. They are indicated by higher soil moisture especially in deeper peat layers and do not form the typical palsa mounds. However, the peat was not completely saturated and infrequently covered with Sphagnum species. The soils at the palsas are characterized as histosols (Kjær et al., 2024). Typical bulk densities of peat plateaus in this region range from 0.08 to 0.28 g cm^{-3} in the active-layer and permafrost peat, with low values below the active layer due to high excess ice contents and low values in the top of the active layer due to high porosity in the surface peat (Kjellman et al., 2018). The site itself is quite heterogeneous with a varying thickness in the peat layer; mineral soil was determined in different depths ranging from 70 to 275 cm. Clayey-silty, frost-sensitive sediments and sandy-gravelly deposits containing embedded layers of sediment-poor massive ice cores can be found below the peat layer (Meier & Thannheiser, 2017). The palsa vegetation on the site consists of *Betula nana, Rubus chamaemorus, Rhododendron tomentosum, Empetrum nigrum, Vaccinium vitis-idaea, V. uliginosum*, *V. myrtilis*, and *Andromeda polifolia, Eriophorum vaginatum* and various lichens (*Cladonia* spp.) located on top of the peat plateaus and the slopes.

The research site included one thermokarst containing several successional stages, namely the recent, young and mature stages. The recent thermokarst stage was characterized by dark brown surface structure that indicated a recently collapsed palsa. This section was

sparsely vegetated with *Eriophorum* species, e.g. *Eriophorum scheuchzeri*, but no sphagnum species have yet accumulated there compared to the other succession stages. In the young thermokarst stage, as well as in the mature, *Sphagnum* and *Eriophorum* species were already growing above the collapsed peat. The young thermokarst was characterized as a highly water saturated section with floating greenish *Sphagnum* species, while the mature area was identified by reddish-brown *Sphagnum* indicating an advanced stage of development.

2.2 Experimental design

Methane fluxes were measured in intact and degraded palsas and in one thermokarst section. In this thesis, only fluxes from the recent stage (as explained under 2.1) were incorporated (Figure 6 c). Degraded palsas were identified by collapsed surfaces in comparison to the intact palsas, which formed mounds that stood out from the rest of the scenery (Figure 6 a). Degraded palsas also seemed to consist of more moist peat.



Figure 6: (A) The Áidejávri research site with intact (mounds) and degraded (depressions) palsas. (B) PVC-collars (B, D, U) that are installed on the palsas. (C) The recent thermokarst transition stage with wooden pallets for enabling flux measurements. The blue pipe in the foreground is a piezometer that was installed for water level measurements.

Three types of collars (Figure 7) made from PVC were used to assess the influence of different peat depths on methane fluxes (see chapter 2.3). They were installed in the beginning of July 2025, as thawing of the active layer has progressed far enough. Undisturbed collars (U) were simply inserted 30 cm into the ground without any further disturbance. The cylindrical shape was open on both ends. Bottom (B) collars were sealed at the end inside the peat to exclude positive methane fluxes from deeper peat layers. B-collars were installed by taking out a peat block with an unsealed collar. The B-collar was inserted into the hole that was created and the removed peat block was put back inside the B-collar inside the hole. Deep (D) collars were installed the same way as B-collars to simulate the disturbance in B-collars. In this way, the effect of excluding fluxes from deeper peat layers could be investigated, as they both had the same disturbance and only differed in the fact that B-collars were sealed at the bottom, while the D-collars were the same shape as U-collars. U-, B- and D-collars were placed close to each other forming one plot group on an intact or degraded palsa (Figure 6 b), which were in total 18 (6x3) on intact palsas and 27 (9x3) on degraded palsas. All of them were the same height (35 cm), the same diameter (23.5 cm) and all were inserted 30 cm into the peat leaving 5 cm above the surface. Several prerequisites had to be met for setting up plots. On intact palsas it was important not to choose spots on slopes; the collars were installed on flat surfaces on top of the palsas only. Other crucial aspects for both intact and degraded palsa spots were avoiding cracks in the peat and only unvegetated, bare peat locations were favored.

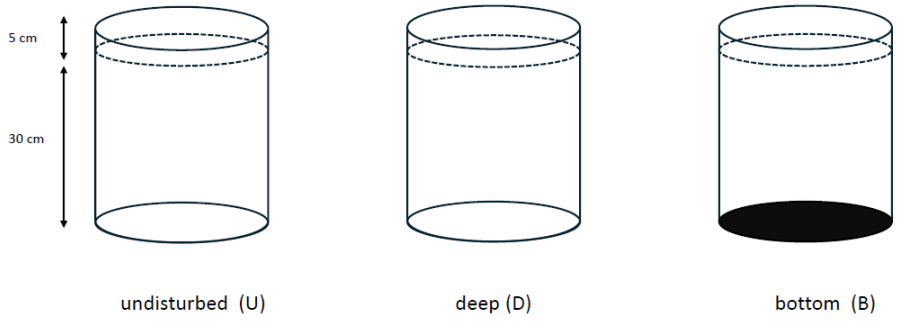


Figure 7: different collar treatments that were installed in the intact and degraded palsas in Áidejávri

Collars used in the recent thermokarst (U-collars only) were already inserted in the first year of the measuring campaign in 2024. However, in this thesis, only thermokarst fluxes from summer 2025 were analyzed. For measuring gas fluxes in the thermokarst, wooden pallets were set up near the collars (Figure 6 c) and put aside after measurements were done.

2.3 Measurement of CH_4 fluxes

Methane fluxes were measured using a portable greenhouse gas analyzer (UGGA) (Los Gatos Research, San Jose, CA, USA). A vented chamber with a volume of 15 L, connected to the gas analyzer with two tubes, was placed on a collar and sealed with a rubber band to prevent gas interchange between headspace and atmosphere (Figure 8). The chamber was equipped with a reflecting foil on the outside so that the temperature inside the chamber would not rise too much and thus impair the fluxes. Tubes were connected to the UGGA and the chamber to pump the gas into the analyzer and then back into the chamber, with a flow rate of ~ 200-300 ml/min. The system was closed for ~ 5 minutes for the measurement until a linear slope developed. The value of the slope was fitted and calculated afterwards using linear regression.



Figure 8: UGGA and mesocosm chamber on an intact palsa plot in Áidejávri

The following equation (4) was used to calculate the methane fluxes (mass flux of methane):

$$F_{CH_4}(mg * m^{-2} * d^{-1}) = \frac{(dc)}{(ct)} * \frac{P_a}{P} * \frac{V_{total}}{R * T * A} * d * 10^3 * M \quad (4)$$

Table 1: Explanation of variables used in equation (4)

$\frac{(dc)}{(ct)}$	flux rate in ppm s^{-1} (slope of a linear regression), divided by 10^6 to convert from ppm per s to $\frac{mol}{mol}$ s^{-1}
P_a	measured air pressure
P	standard air pressure (1 013 hPa), used to convert to unit atm
V_{total}	total volume in the mesocosm chamber (V = 0.015 m^3), (0.0001 m^3 was added for gas volume in the tubes)
R	gas constant (8.2057 * 10^{-5} $m^3 atm\, mol^{-1} K^{-1}$)
T	air temperature inside the chamber in Kelvin
A	chamber base (0.043 m^2)

d	1 day in s (86 400 s) for conversion of flux per s into flux per day
10^3	for conversion of g into mg
M	molar mass of CH_4 (16.0425 g * mol^{-1})

2.4 Incubation Experiment

An incubation experiment was carried out to investigate how methane fluxes vary between peat layers at the same temperature. Peat cores were sampled from intact (n = 36) and degraded palsas (n = 42) at a distance of 1-2 m from the collars at the research site on the 25th of August 2025. For this purpose, a stainless-steel tube (diameter 8.02 cm) was inserted in 10 cm increments on intact and degraded palsas ranging from 0-10 cm up to 60-70 cm. However, not every core reached this depth, because some were limited by the permafrost ice and sampling stopped when reaching the ice layer.

Afterwards, the samples were pre-incubated at room temperature overnight inside a plastic bag in which they were carried from the field and then measured for the next two days. Before measurements started, the samples were stored in a cylindrical glass container (V = 0.0010502 m^3) without a lid and put in a water bath, to set them to 15 °C. For flux measurements, the samples had a temperature of ~ 15 °C and were taken out of the water bath. The glass container was then sealed with a metal lid which was prepared with two connections for the plastic pipes of the greenhouse gas analyzer. Each sample was measured once for approximately 10 minutes.

For flux calculation, the formula used above (formula 4 in chapter 2.3) was adapted to fit the incubation experiment. The flux unit used for the incubation was µg kg^{-1} h^{-1} instead of mg m^{-2} d^{-1}. For the total volume (V_{total}), the remaining volume in the glass container was calculated ($V_{glass} – V_{sample}$). Instead of air temperature, the temperature of the peat sample was used. Furthermore, the flux was related to the dry weight of the peat sample instead of a chamber base area. The flux is also calculated per hour instead of a daily flux rate.

After flux measurements, living roots were sorted out and then dried at 60 °C to calculate the root biomass per unit area for each depth. Additionally, the gravimetric water content was determined to assess the dry weight of each sample. Aliquots of the samples were dried at 60 °C for 24 hours until mass constancy.

2.5 Measurement of soil physical parameters

To obtain the volumetric water content θ (VWC), firstly the soil moisture (in mV) was measured in the field after each flux measurement with a soil moisture probe (ThetaProbe ML2x) (Delta-T Devices Ltd., 1999), which was inserted into the soil three times at different points inside the collar to account for heterogeneity. Secondly, a soil moisture calibration was conducted using

six soil samples of defined identical volume. These samples then were saturated with water and the weight as well as the soil moisture (in mV) was noted. The samples then were air-dried, and soil moisture was measured in regular intervals until weight constancy. For each value measured in mV, the volumetric water content was calculated by using the formula below:

$$\theta = \frac{volume\ of\ water}{total\ soil\ volume}\ [\%] \tag{5}$$

Afterwards, polynomial functions of degree three (Figure 9) were fitted to display the correlation between soil moisture (x-axis) and volumetric water content (y-axis).

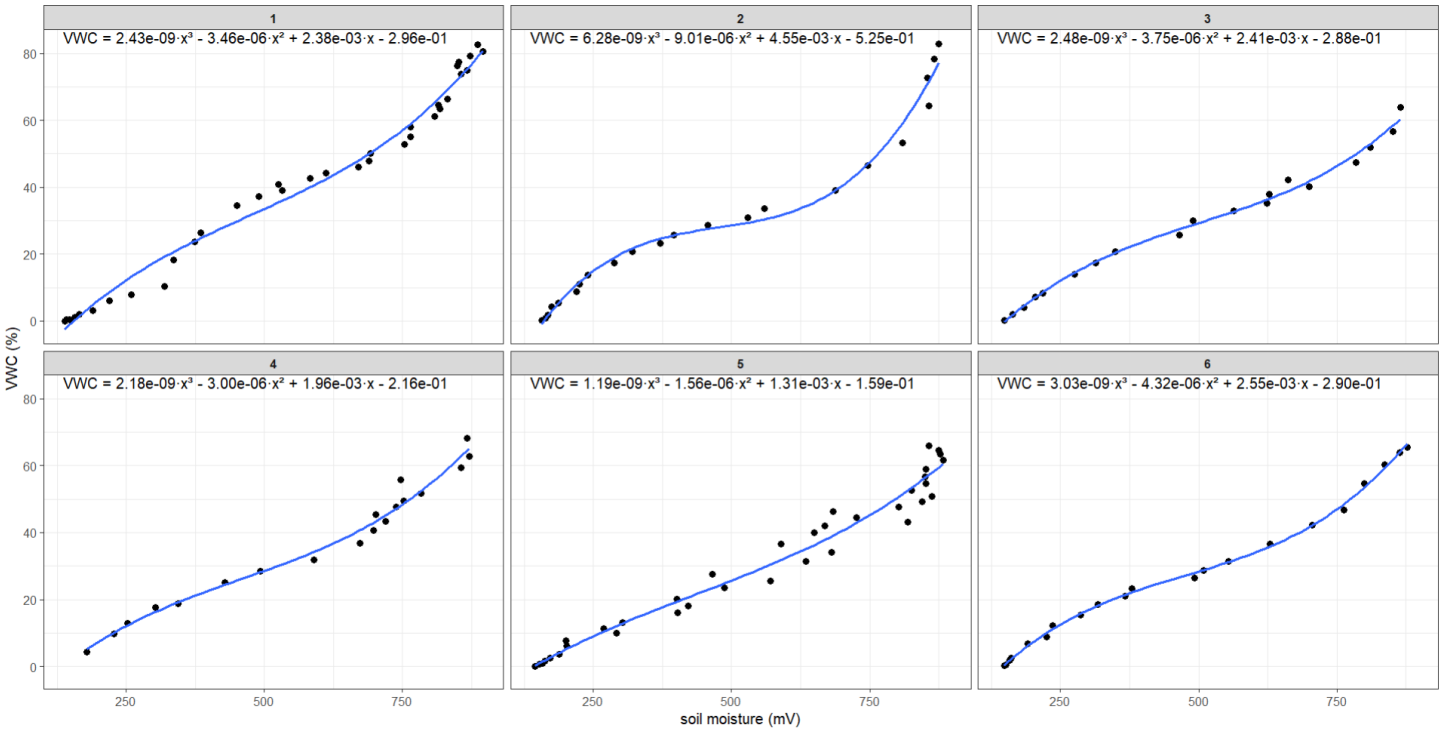


Figure 9: polynomial functions of degree three that were fitted to calculate the volumetric water content from the soil moisture values in mV measured in the field

Each of the derived functions was used to calculate the VWC (in %), thus resulting in six VWC-values for each in-situ measured soil moisture. A mean value was calculated for the final volumetric water content.

Soil temperature was measured with a soil temperature probe 10 cm below ground level after flux measurements to prevent impairing an influence on the fluxes. Temperature was measured two times inside each collar, keeping the temperature sensor inside the soil for three minutes each so that the correct temperature could adapt.

Thaw depth was measured with a thaw depth probe that was inserted in a 90 ° angle into the ground until the ice layer was reached. Afterwards the length of the probe was measured from the bottom to the point where the pole was at the same height as the ground level. At each plot location (place where three collars were next to each other) the thaw depth was measured two times in unvegetated and once in vegetated peat.

2.6 Statistical analysis

All statistical analysis was done using R (version 4.5.0) (R Core Team, 2025).

In general, linear mixed effects models (LMEM) combined with pairwise comparisons from the emmeans package (Lenth & Piaskowski, 2025) were applied to test for significant differences in fluxes among treatment groups. Models were fitted using the glmmTMB package, allowing for heteroscedasticity (Brooks et al., 2017; McGillycuddy et al., 2025). Thaw stage (intact / degraded palsa or recent thermokarst) was included as fixed effects while collar-ID and date were selected as random effects. Model diagnostics were performed using DHARMa package (Hartig, 2024) and to check whether adaptations are necessary. To meet model assumptions (uniform distribution of residuals), data were transformed using the Yeo-Johnson transformation (Yeo & Johnson, 2000), implemented in the bestNormalize package (Peterson, 2021; Peterson & Cavanaugh, 2020). Additional LMEMs were used to assess the influence of environmental parameters (soil temperature, VWC, thaw depth) on flux rates. As effects of environmental parameters did not differ between palsa type, additive models without interaction terms were applied. Prior to model analysis, predictor variables were z-standardized using the scale-function (sample mean = 0, SD (standard deviation) = 1) (Schielzeth, 2010) to allow direct comparison of effect sizes across variables despite differing measurement units of environmental drivers of CH_4 fluxes. In case a Yeo–Johnson transformation was applied, effect sizes (estimates) were reported on the transformed scale since they cannot be directly interpreted in original units.

For **hypothesis 1**, the LMEM was used to compare fluxes from intact and degraded palsas and recent thermokarst. Arithmetic means from raw data were calculated to represent average flux rates across the entire measurement period. To identify the influence of environmental variables (temperature, VWC, thaw depth), another LMEM was utilized, where the Yeo-Johnson transformation was applied to the data beforehand. The same LMEM was performed with the thermokarst data, using soil temperature, VWC and water level as predictor variables to test their influence and intensity on the response variable

For **hypothesis 2**, no additional linear mixed effects model was set up, the influence of environmental parameters was all tested with one model, but the results of the influence of thaw depth on methane fluxes were only used in hypothesis 2.

To test **hypothesis 3 a**, a Principal Component Analysis (PCA) was conducted to indicate the strength of contribution of environmental drivers to the fluxes. The prcomp command from the stats package (R Core Team, 2025) was used to calculate the PCA. Visualization was done using the ggplot2 package (Wickham, 2016). After the PCA, a Pearson-Correlation was conducted to see which environmental parameter had a significant (significance level = 0.05) influence on the fluxes.

For **hypothesis 3 b**, an LMEM was used to test significant differences between bottom and deep collars on intact and degraded palsas. Besides, an LMEM was used to see whether the environmental variables temperature and VWC affected flux rates significantly. To illustrate

whether fluxes from bottom collars had higher methane uptake, the difference between bottom and deep collars was calculated and graphically visualized.

All environmental parameters (thaw depth, soil temperature, VWC) were tested whether they were significantly different in intact and degraded palsas. For this purpose, an LMEM and pair-wise comparison (from the emmeans command) was used for each environmental parameter with collar-ID and date as random effects and palsa type (intact / degraded) as fixed effect. In some cases, the data was transformed using logarithms.

3 Results

3.1 CH_4 fluxes in the palsa – thermokarst structure

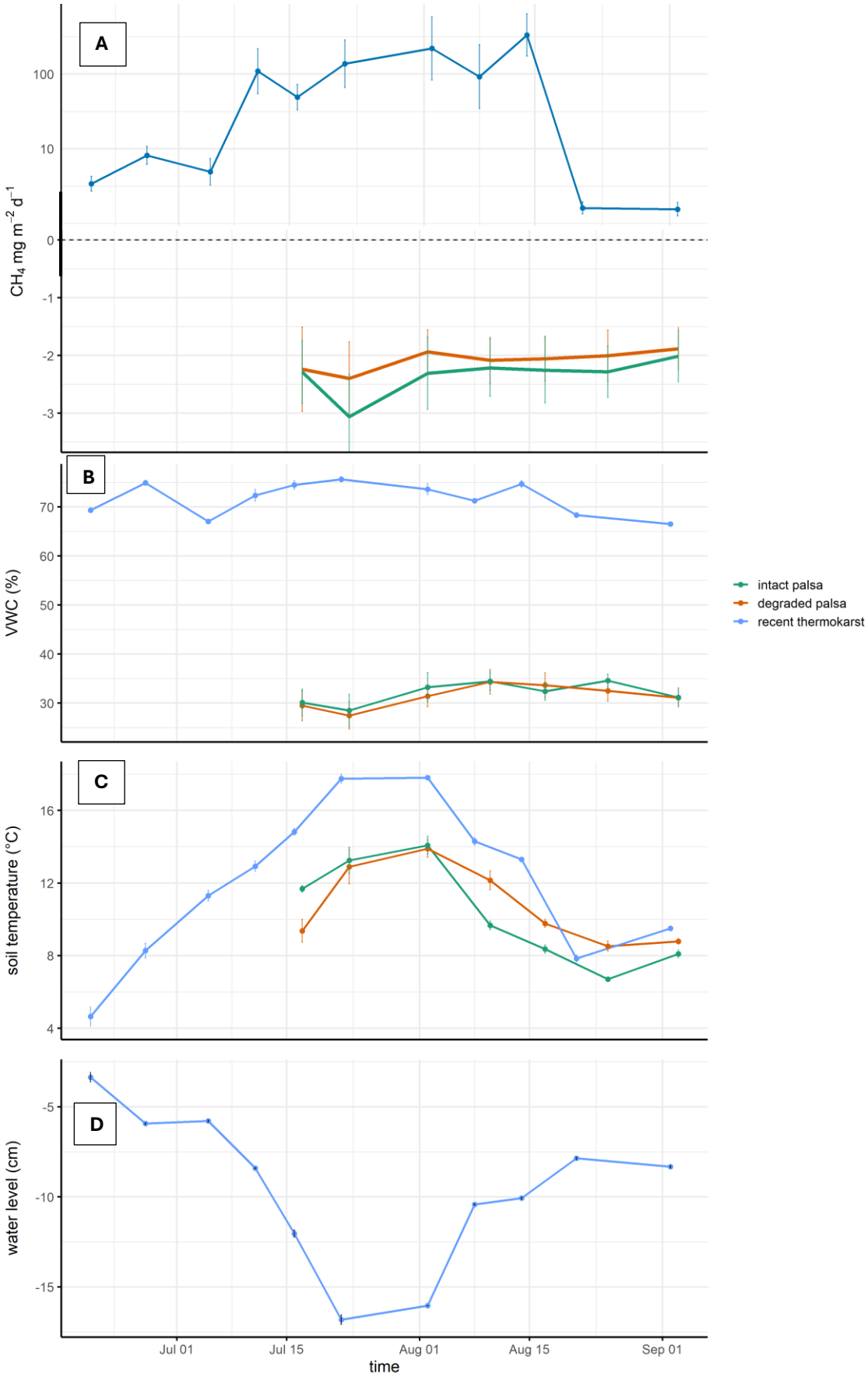


Figure 10: (A) CH_4 -fluxes in intact and degraded palsas and recent thermokarst. As fluxes from thermokarst and palsas extremely differed in their magnitude and would otherwise not fit into one diagram, values from the recent thermokarst were log-transformed, while palsa flux rates remain unaffected. (B - D) environmental parameters that are related to the fluxes, calculated as daily averages (D) The water level is measured in cm below ground level.

The intact palsas were taking up slightly more methane (-2.35 mg CH_4 m^{-2} d^{-1}) than the degraded palsas (-2.09 mg CH_4 m^{-2} d^{-1}) during the period, but the difference was not significantly different (Table 2). During the short period between mid-July and beginning of September, there were only minor fluctuations in methane flux rates (Figure 10 a). Significant differences were only found when comparing fluxes from both palsas with the higher thermokarst fluxes (both $p < 0.001$). The recent thermokarst acted as a net methane source (80.7 mg CH_4 m^{-2} d^{-1}) throughout the season. Highest fluxes were found mid-season from mid-July to mid-August (maximum single flux rate: 866.7 mg CH_4 m^{-2} d^{-1}) and dropped rapidly after mid-August to 0.21 mg CH_4 m^{-2} d^{-1} (minimum single measured flux) (Figure 10 a).

Mean volumetric water contents were not different between intact (32.1 %) and degraded (31.4 %) palsas. In contrast, the water content in the thermokarst showed consistently high results (71.6 %), indicating no seasonal trend but revealing a significant difference compared to both palsa types ($p < 0.001$ for comparison with intact and degraded palsas). A seasonal trend is indicated by the soil temperature, peaking at the beginning of August in both the intact and degraded palsas (Figure 10 c). While the temperature was lower in the degraded than in the intact palsas until beginning of August, an opposite pattern was observed thereafter. Over the total period, soil temperature was slightly higher in the degraded palsa (Figure 10 c). Mean soil temperatures in the thermokarst (12.01°C) were significantly higher compared to the palsas (intact: $p < 0.001$; degraded: $p = 0.0021$). Soil temperature in the thermokarst behaved contrary to the water level (9.5 cm), since water levels were lowest when temperatures reached the highest values (Figure 10 c and d).

Table 2: Estimate, standard error and p-value (significance level 0,05) of the comparison of the transition stages intact palsa, degraded palsa and recent thermokarst based on the Yeo-Johnson transformed model values.

	estimate	SE	p-value
intact palsa – degraded palsa	-0.107	0.330	n.s.
intact palsa – thermokarst	-1.565	0.415	< 0.001
degraded palsa – thermokarst	-1.458	0.390	< 0.001

Table 3 illustrates the effects of soil physical parameters on methane fluxes from intact and degraded palsas. The estimates suggest that thaw depth impairs methane uptake, while soil temperature enhances it, but neither of these predictors affected the response (flux) significantly, as already indicated graphically in Figure 10. An evaluation of the random effects revealed higher variation across collars (1.3) than across date (0.0004), indicating overall heterogeneity in flux data from the site.

Table 3: influence of soil physical parameters on intact and degraded palsas as well as from recent thermokarst, indicated by estimates, SE (based on the estimate) and p-value.

	sites	estimate	SE	p-value
thaw depth	palsa (intact + degraded)	0.023	0.078	n.s.
soil temperature	palsa (intact + degraded)	-0.070	0.053	n.s.
volumetric water content	palsa (intact + degraded)	-0.034	0.110	n.s.
VWC	recent thermokarst	0.508	0.121	< 0.001
soil temperature	recent thermokarst	0.683	0.269	0.011
water level	recent thermokarst	-0.476	0.265	n.s.

The thermokarst fluxes were significantly influenced by soil temperature (p = 0.011) and VWC (p = < 0.001). Water level indicated no significant influence on fluxes (Table 3). Soil temperature had the strongest effect among the other predictors in the thermokarst (0.683), closely followed by VWC (0.508). Due to the non-linear transformation, effect sizes are reported on the transformed scale. Variation between collars ($9.22*10^{-2}$) was higher than variation between days ($5.23 *10^{-10}$).

3.2 Influence of thaw depth on methane fluxes in intact and degraded palsas

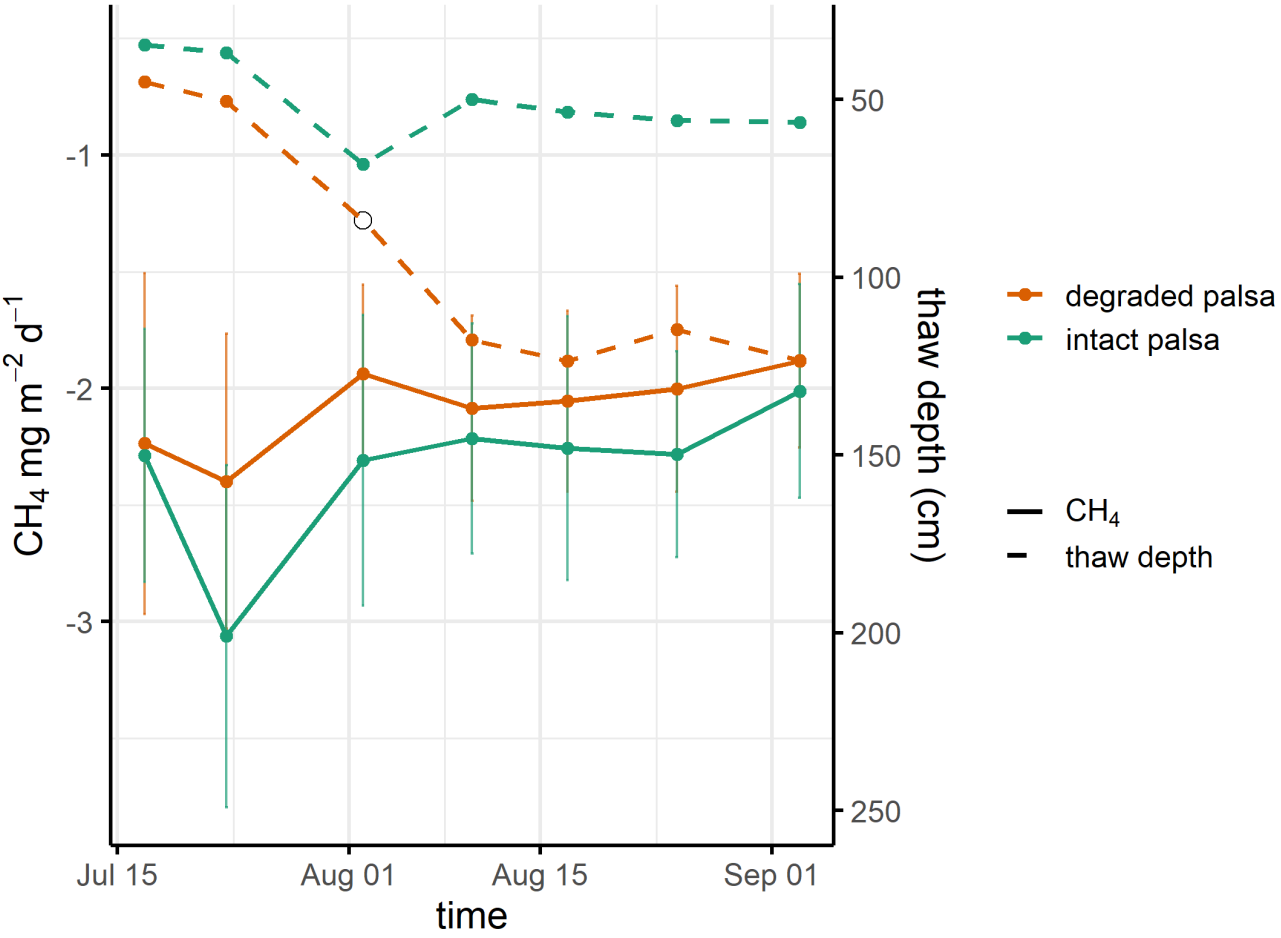


Figure 11: thaw depth and methane fluxes throughout the measuring period. The left axis displays methane uptake; the right indicates thaw depth. The white data point in the thaw depth graph from degraded palsas shows an interpolated value due to a missing measured value on the 2nd of August 2025.

Thaw depth on intact palsas started at approximately 35 cm on average measured on 17.07.2025, whereas on degraded palsas thaw depth was already averaging 45 cm. At the end of the measuring period (03.09.2025), mean thaw depths reached 123 cm in degraded palsas, and 57 cm in intact palsas. Thaw depths were significantly different in intact and degraded palsas ($p < 0.001$). Over the season, methane fluxes did not increase with increasing thaw depth in both palsa structures.

3.3 Incubation of peat samples from consecutive deeper layers

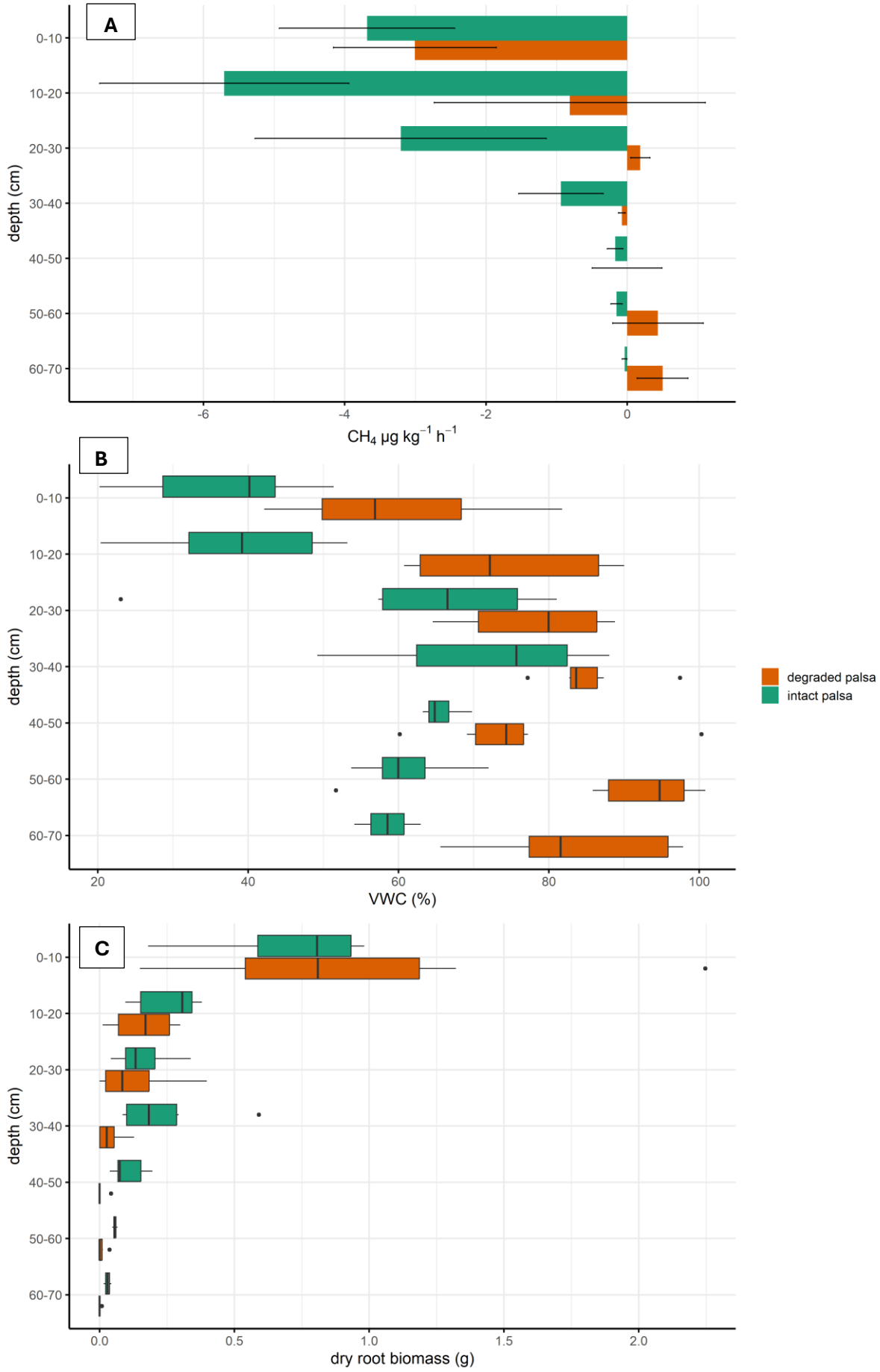


Figure 12: (A) relative CH_4 fluxes (mean) and SE from peat samples taken from intact and degraded palsas, originating from consecutive deeper peat layers (B) Volumetric water content (VWC) with SE and (C) root biomass with SE from peat samples in intact and degraded palsas

Methane uptake from intact palsa samples was highest in samples at 10-20 cm depth and gradually declined to almost zero with increasing depth (Figure 12 a). All peat depths from the intact palsa showed consistently methane uptake. The peat samples from degraded palsas behaved differently, since the highest uptake of methane was in the upper layer (0-10 cm), but this potential uptake is still below the level of the highest uptake from intact palsa samples. The mean potential uptake from the next deeper layer (10-20 cm) is on average already less than half the uptake from the previous layer above. With increasing depth, methane uptake is limited, especially at 40-50 cm depth where fluxes varied from -1.7 to 1.9 µg kg^{-1} h^{-1}. Eventually, in deeper layers positive mean values were measured; highest values occurred in samples from the 50-60 cm and 60-70 cm layer.

The median volumetric water content (VWC) was continuously higher in the samples from degraded palsas than in those from intact palsas (Figure 12 b). Degraded palsa samples imply a higher variation in VWC in 60-70 cm depth. In 30-40 cm depth, samples from intact palsas suggest a higher variation. Tendentially, the spread of values declines towards deeper soil layers in samples from intact palsas. Depth and VWC were positively correlated (samples from intact palsas: $r = 0.54$, $p = 0.001$, samples from degraded palsas: $r = 0.46$, $p = 0.002$), implying that water content increases with increasing depth.

Root biomass was in general slightly higher in intact palsa peat samples (Figure 12 c), except in the upper peat layer (0-10 cm), where median values were similar, but with a wider range observed in degraded palsa samples. The amount of root biomass decreased with increasing depth in both intact and degraded palsa samples. This reflects in a significantly negative correlation of root biomass and depth in samples from intact and degraded palsas ($r_{intact} = -0.67$, $r_{degraded} = -0.57$, both $p < 0.05$).

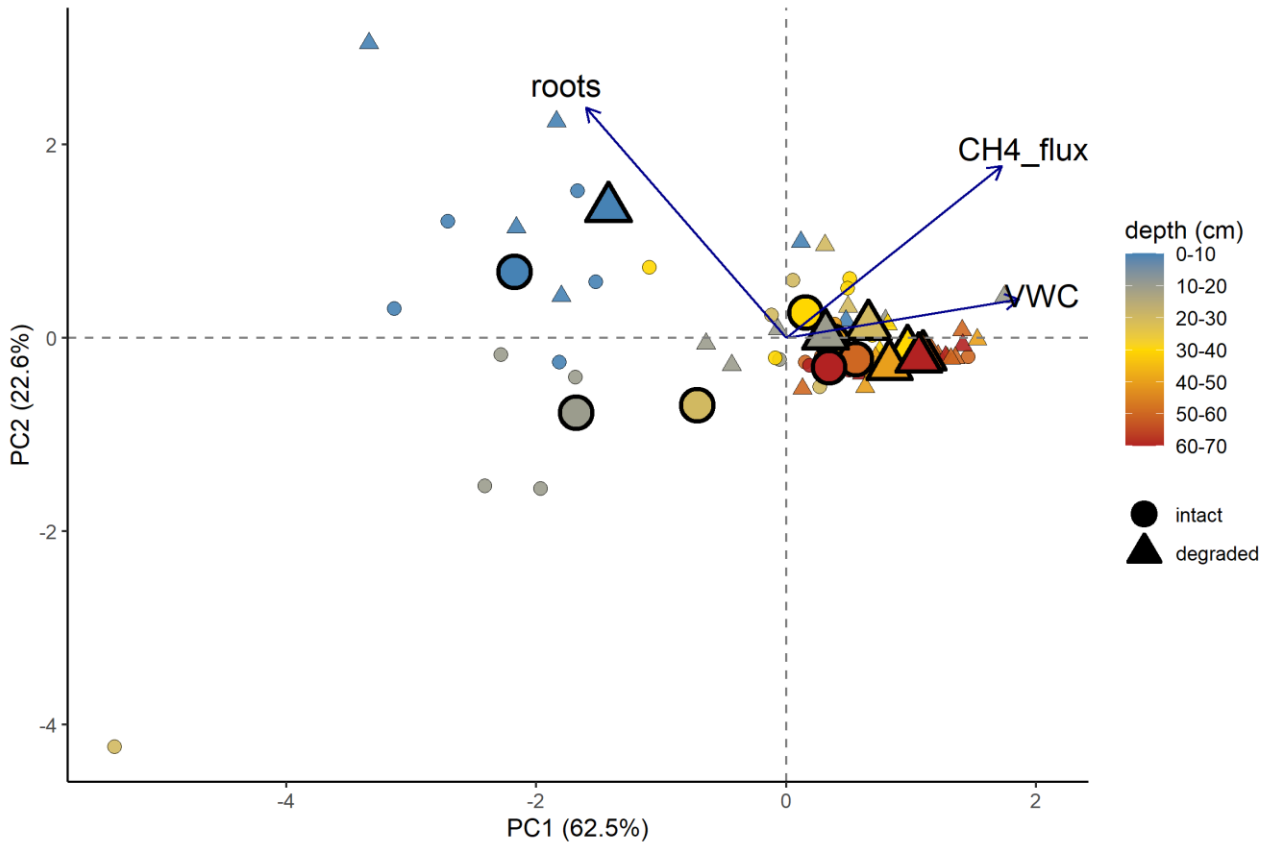


Figure 13: The Principal Component Analysis (PCA) visualizes which variables (roots, volumetric water content, depth gradient) influenced the fluxes in the incubation experiment. Shapes of data points represent measurements derived from intact (circle) or degraded (triangle) palsa peat samples. Values from shallow samples are indicated by a bluish to beige color, values from deeper samples are colored rather orange to red. Bold data points represent the average PCA score of each group, indicating the center of the data in each group

The first aspect to look at is the amount of data variance since the PCA sorts the data variance into the major features on the leading dimensions and what can be considered random noise on the minor dimensions (Greenacre et al., 2022). The major features are represented by the first consecutive dimensions: PC (principal component) 1 already explains 62.5 % of the variance, and together with PC 2 (22.6 %), 85.1 % of the variance can be explained (Figure 13). Only 14.9% of the variance is declared random or unexplained. Vectors that are pointing in the same direction suggest that these soil physical parameters are strongly correlated, conversely, when pointing in the opposite direction, this suggests a negative correlation. A positive correlation is indicated for fluxes and VWC (Figure 13), which intends that increasing values in VWC are related with increased fluxes and thus an inhibited methane uptake. Vectors that are perpendicular to one another are suggested not to be correlated. This is the case for methane fluxes and root biomass.

Symbols along the VWC vector are predominantly orange and red and triangular, suggesting that degraded palsa samples and deeper layers are associated with higher VWC. In contrast, the root biomass vector is associated with blue and grey symbols, indicating shallower peat layers. Symbols are systematically arranged along PC1, with near-surface symbols

(blue/grey) at lower values and symbols indicating deeper samples (red/orange) at higher values, hence reflecting a depth-related gradient. Methane flux and VWC vectors point in the direction of increasing PC1, indicating higher volumetric water content and methane fluxes with increasing depth. Conversely, the arrow representing root biomass points to the lower end of the PC1 axis, indicating higher root abundance in shallower peat layers.

The Pearson-Correlation conducted afterwards is intended to test whether the results from the PCA reflect in the derived p-values (Table 4). As suggested by the results, root biomass was significantly negatively correlated with the fluxes from peat samples originating from intact ($r = -0.38$, $p = 0.025$) and degraded ($r = -0.53$, $p = < 0.001$) palsas. However, the PCA did not indicate a relationship between both variables, as the vector of roots is located perpendicular to the vector representing the flux.

Table 4: p-values and correlation coefficients derived from the Pearson Correlation, where the influence of VWC, root biomass and depth gradient was tested for significance on intact and degraded palsa peat samples.

	intact palsa	**degraded palsa**
VWC p-value	0.004	<0.001
VWC correlation coefficient	0.48	0.53
root biomass p-value	0.025	<0.001
root biomass correlation coefficient	-0.38	-0.53
depth gradient p-value	0.002	0.011
depth gradient correlation coefficient	0.51	0.39

Depth also was a significant driver determining fluxes both on samples from intact ($p = 0.002$) and degraded ($p = 0.011$) palsa peat samples. As depth increases, fluxes tend to rise (intact: $r = 0.51$, degraded: $r = 0.39$), resulting in limited methane oxidation. Likewise, the VWC was positively correlated with fluxes.

3.4 Methane fluxes from sub peat layers

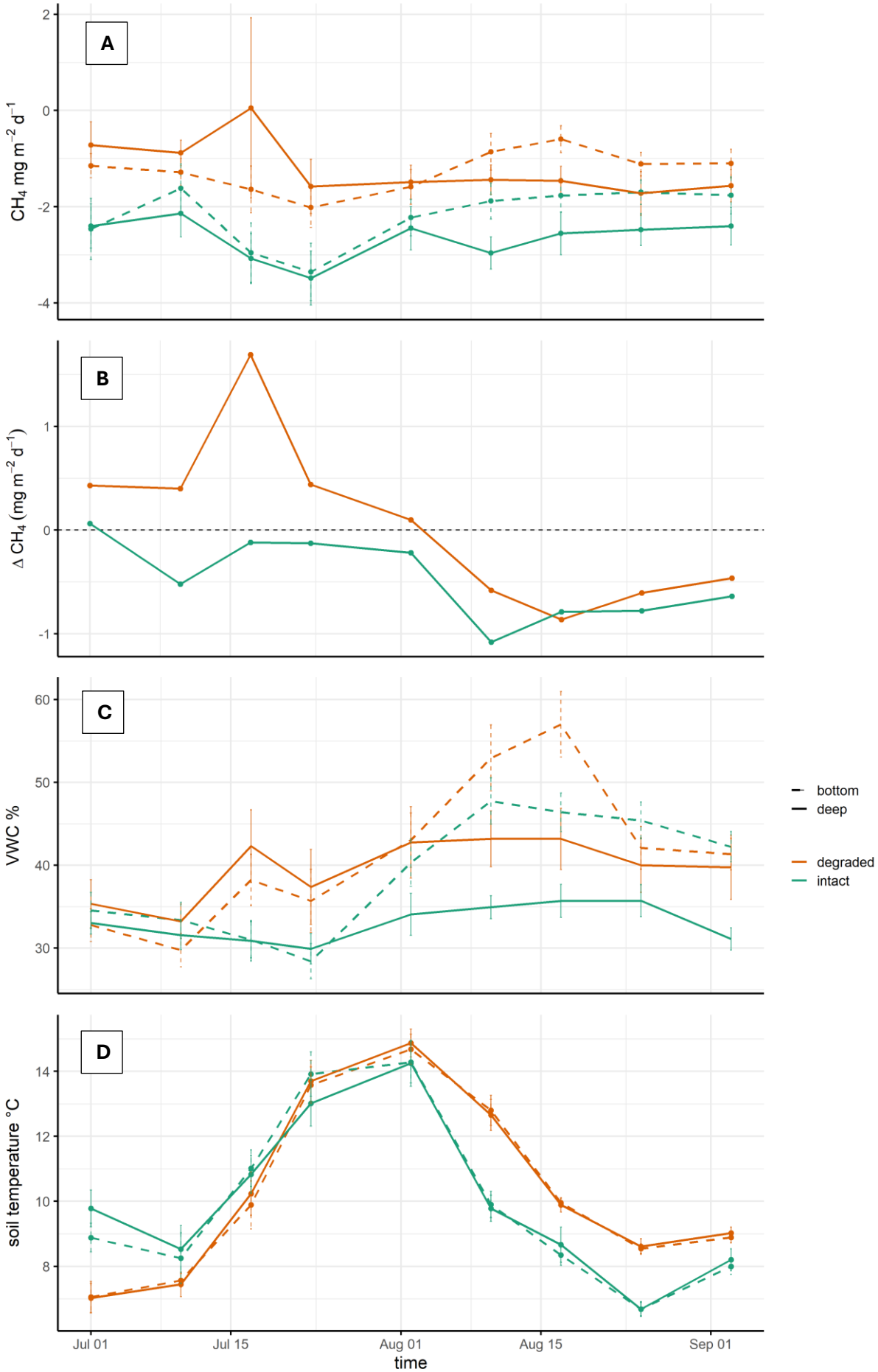


Figure 14: (A) CH_4 fluxes in intact and degraded palsas over the growing season with two treatment types each – bottom collars are displayed with a dashed line, and deep collars are indicated by a solid line. (B) flux rates from bottom collars were subtracted from flux rates from deep collars (deep – bottom). Positive values (above the black dashed line) indicate higher fluxes from bottom collars. (C) and (D) VWC and soil temperature measured in bottom and deep collars on intact and degraded palsas during the growing season

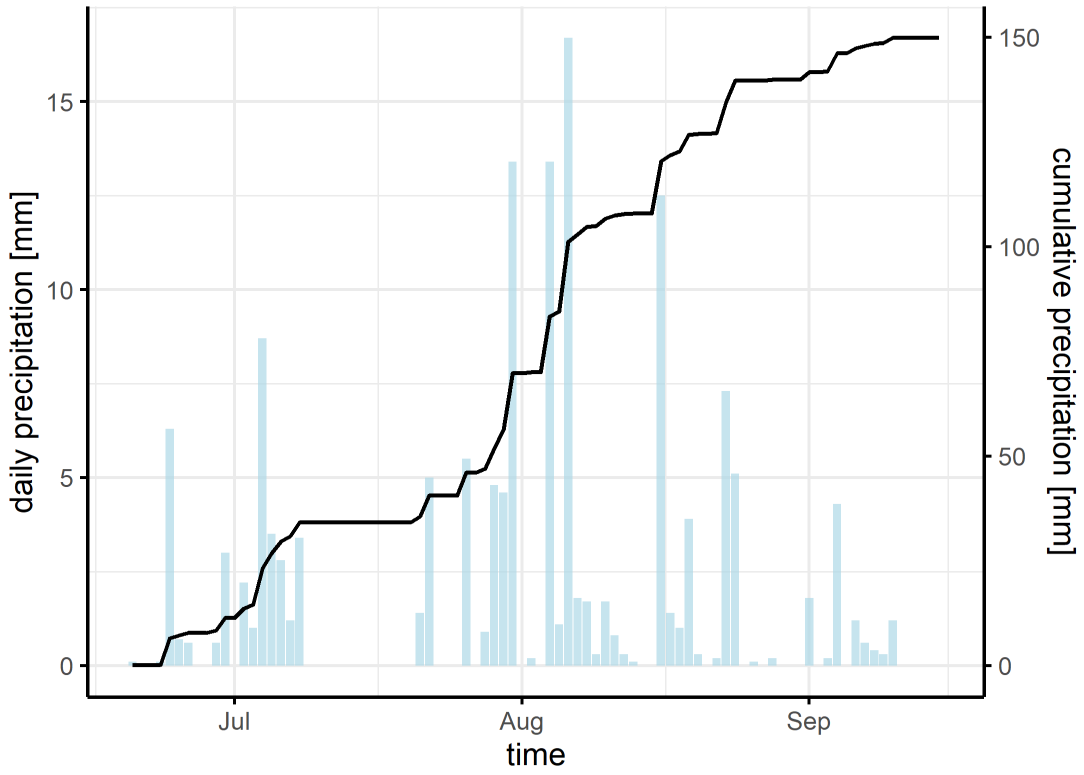


Figure 15: Precipitation at Áidejávri during the period. The daily precipitation (in mm) is featured on the left axis, while the cumulative precipitation (mm) is indicated by the right axis. Precipitation data is taken from the Norwegian Meteorological Institute measured at the Kautokeino climate station.

Figure 14 a shows seasonal mean fluxes from deep (intact: -2.66 mg CH_4 m^{-2} d^{-1}, degraded: -1.20 mg CH_4 m^{-2} d^{-1}) and bottom (intact: -2.19 mg CH_4 m^{-2} d^{-1}, degraded: -1.26 mg CH_4 m^{-2} d^{-1}) collars on intact and degraded palsas. Fluxes measured on intact palsas were constantly below the level of degraded palsas, indicating a higher uptake on intact palsas. When comparing fluxes from bottom and deep collars on each palsa type, deep collar fluxes are most of the time characterized by a higher uptake. This issue is illustrated in Figure 14 b, where the difference between deep and bottom collars was calculated; values above the black dashed line highlight higher methane oxidation from bottom collars. Higher B-collar uptake was only the case on degraded palsas from the beginning of the measurement period until the beginning of August. On intact palsas, methane uptake was always higher in deep collars during the whole time of measurement period.

As indicated in Table 5, fluxes from deep and bottom collars did not differ significantly both in intact and degraded palsas. The positive estimates suggest that fluxes in deep collars were linked to higher methane uptake. Methane uptake from deep collars on intact palsas is on average 0.47 mg CH_4 m^{-2} d^{-1} higher than from bottom collars, while on degraded palsas the deep collar uptake is 0.06 mg CH_4 m^{-2} d^{-1} greater.

As already observed in U-collars in palsas and in the thermokarst (Figure 10), soil temperature followed a seasonal trend in bottom- and deep-collars (Figure 14 d), peaking at the beginning of August. Around mid-July, the temperature in degraded palsas (averages: bottom: 10.3 °C, deep: 10.4 °C) rose above the temperature in intact palsas (bottom: 9.9 °C, deep: 10.0 °C) and remained above temperatures in intact palsas.

In the beginning of the period, VWC, as illustrated in Figure 14 c, remained on a lower level in all plots, and started to increase between mid-July and the beginning of August. B-collars reached higher levels (averages: intact 38.8 %, degraded 41.4 %) than D-collars (intact: 33.0 %, degraded: 40.0 %) and remained in major positions until the end of measurements at the beginning of September. Yet differences between VWC from deep and bottom collars were not significant in intact and degraded palsa treatments. As featured in Figure 15, precipitation was highest in August, peaking at roughly 17 mm. The time before and after is characterized by days with less rain, yet at regular intervals. The period of higher VWC in bottom collars follows the time of higher precipitation.

Table 5: estimates, SE and p-values for (i) differences between bottom and deep collars and (ii) the effects of VWC and soil temperature on fluxes from deep and bottom collars in intact and degraded palsas.

	estimate	SE	p-value
intact bottom – intact deep	0.47	0.57	n.s.
degraded bottom – degraded deep	0.06	0.50	n.s.
VWC	0.43	0.07	< 0.001
soil temperature	0.10	0.09	n.s.

The effect of volumetric water content on the fluxes in intact and degraded palsas both in deep and bottom collars was highly significant ($p < 0.001$), indicating that a 1 SD (standard deviation) change in VWC (SD_{VWC}= 10.4) leads to a reduction in methane uptake by 0.43 mg CH_4 m^{-2} d^{-1} (Table 5). The effect of soil temperature was not significant. As predictors were z-standardized prior to analysis, estimates serve as effect size and are directly comparable, indicating that VWC had a 4 times stronger impact on methane uptake than soil temperature.

4 Discussion

4.1 CH_4 fluxes in a palsa – thermokarst peatland

In the first hypothesis it was expected that methane fluxes alter depending on the transitional stages, ranging from a sink in intact and degraded palsas to a source in the recent thermokarst structure. Average fluxes were -2.35 and -2.09 mg CH_4 m^{-2} d^{-1} in intact and degraded palsas, respectively, indicating that the disturbance regime in degraded palsas had a negligible, non-significant effect on the sink strength for atmospheric methane. However, hypothesis 1 has been partly confirmed as the recent thermokarst exhibited a methane source with an average of 80.7 mg CH_4 m^{-2} d^{-1}.

Methane oxidating bacteria showed a higher activity in intact palsas since more atmospheric methane was assimilated than in degraded palsas (Figure 10 b). These methanotrophs convert methane to carbon dioxide, acting as a net sink for atmospheric methane in well aerated soils. As they are dependent on oxygen supply, limited diffusion of O_2 into the soil, determined by soil moisture content, can reduce the CH_4 uptake rate (Juncher Jørgensen et al., 2024; Kolb, 2011). However, non-significant differences between intact and degraded palsa methane fluxes and VWC (chapter 3.1) indicate that the disturbance of the microbial community in degraded palsa and the formation of anaerobic zones remain insufficient, despite degraded palsa samples showed slightly higher VWC values at depth (chapter 4.3). These VWC measurements, however, represent only a temporal snapshot. In addition, degraded palsas exhibited a deeper thaw depth, which could theoretically enhance methane production potential in deeper peat layers. Chen et al. (2025) confirmed, that permafrost degradation is associated with a deepening of the active layer, represented by differing median thaw depths at the research site in this thesis (49 cm for intact versus 108 cm for degraded palsas). The same study emphasized that active layer deepening can expand anaerobic soil volumes and reduce the methanotrophic oxidation zone, thereby increasing the potential for methane production. However, as degraded palsas did not show signs of limitation in CH_4 oxidation, it is suggested that methane uptake remained effective despite these physical changes. Potential microbial shifts, as stimulation of methane producing bacteria within the transition zone, likely occur gradually, as communities respond to temperature and thaw alterations (Johnston et al., 2019) but prevailing methane oxidation processes appear sufficient to counterbalance emerging methanogenic activity. Furthermore, these findings suggest that degraded palsas currently represent an early stage of transition, in which conditions favorable for methanogenesis are emerging but microbial communities have not yet fully responded. This so-called lag time is reported by several studies (Holm et al., 2020; Treat et al., 2015). Consequently, although methanogenic communities already exist in the peat, increased CH_4 production and thus limited methane oxidation may emerge only in advanced degradation stages under favorable thermal and redox conditions (Treat et al., 2015).

None of the predictors (soil temperature, VWC) influenced the fluxes in palsas significantly (Table 3). Similar findings were made by Juncher Jørgensen et al. (2024), where no correlation was found between the VWC in the topsoil (0-5 cm) and a large variability of net CH_4 fluxes for the same VWC. In their study, the same applies to the soil temperature in depths of 2.5 and 10 cm. Similar relationships were revealed in this dataset (Figure 25). The authors of the study explain this observation with the scheduling of measuring in situ CH_4 fluxes during the growing season and the seasonal drying of the topsoil where CH_4 oxidation is therefore expected to be limited. In their experiment, highest methane oxidation was seen in depths of 5 to 15 cm and is suggested to be lessened in the 0 to 5 cm interval. Similar findings are indicated in chapter 4.3 in intact palsa samples, where potential oxidation was highest in 10-20 cm depth. Hence, the optimal zone for CH_4 uptake presumably shifts downward the soil profile over the season pursuing an optimal water content, which results in a mismatch between the actual placement of soil moisture measurements (0 – ~ 6 cm) and zone of maximum CH_4 oxidation. This might hint at a decoupling between the topsoil and soil CH_4 oxidation metabolism and suggest that the net surface flux is rather controlled by CH_4 oxidation processes in deeper soil layers than by the soil physical conditions directly in the top layer (Juncher Jørgensen et al., 2024). A missing correlation between soil temperature and CH_4 flux might also be since the temperature variation was too low to affect fluxes significantly. In situ temperatures ranged from 6 to 18 °C in the measurement period, while significant differences due to temperature increase in in situ conditions are often only observed over a longer period (e.g. several seasons) or in controlled incubation experiments (Voigt et al., 2023). For instance, Voigt et al. (2023) only found statistically significant differences between methane oxidation in soil samples tempered at 4 °C and 20 °C in dry soil conditions. The authors implied that soil temperature acts as a secondary control only if substrate (e. g. nutrient availability: dissolved N and labile C) and moisture conditions are favorable for methanotrophs, with moisture being the most influential driver inhibiting methanotrophy. Their findings further suggest that other important controls (e.g. biotic controls) often override the effect of temperature in in situ experiments (Juncher Jørgensen et al., 2024; Voigt et al., 2023). The weak temperature response may also be attributable to limited methane availability, resulting from diffusion barriers within the soil and the low atmospheric methane concentration (Smith et al., 2018) and limited methane production from above the permafrost zone due to scarce amounts of labile carbon necessary for methanogens and highly humified organic material in the palsas (Baysinger et al., 2025; Hodgkins et al., 2014). Consequently, the reaction rate cannot be substantially increased due to insufficient amounts of CH_4 (Smith et al., 2018).

That the thermokarst represents a net source of methane is attributable to water saturation below 3 – 17 cm and constantly high soil moisture above the saturated zone, favoring anaerobic conditions (Figure 10 c). Oxygen diffusion is minimized 10,000-fold in water compared to in air (Somerville & Proctor, 2013), consequently water table and soil moisture, the latter being identified a significant driver for thermokarst CH_4 emissions, control the oxic-anaerobic boundary

and the aerobic or anaerobic metabolic pathway of microorganisms (Ingram, 1983). Anaerobic state is one prerequisite for methane production, in addition to other conditions such as a low redox potential (Conrad, 2020) and absence of alternative electron acceptors such as nitrate, ferric iron and sulfate (Knoblauch et al., 2018). Other anaerobic redox processes yield higher amounts of energy and are therefore thermodynamically favored by microorganisms until these electron acceptors are depleted, after which methanogenesis can commence (Knorr & Blodau, 2009). Iron and Sulphur analyses (Henrich, 2025) from two peat cores in an adjacent *Sphagnum*-covered thermokarst in Áidejávri (up to 175 cm and 250 cm depth, respectively) show low near-surface concentrations (< 2.5 mg Fe g^{-1} and < 2 mg S g^{-1}). With depth, S increased to 7 mg S g^{-1} at 100 cm and 150 cm, respectively and then remained relatively stable at 2-3 mg S g^{-1} below this depth. Fe showed different patterns in the two cores, with one increasing gradually to 8 mg Fe g^{-1} at 225 cm and the other showed peaks at 10 and 50 cm depth (7 mg Fe g^{-1}) (Henrich, 2025). Because inorganic electron acceptors were lowest near the surface, this zone would be expected to least inhibit methanogenesis. Consistent with this, Knorr & Blodau (2009) reported methane production starting while ferric iron and sulfate were still being reduced, suggesting that methanogenesis occurred in micro-environments, especially in the rooted surface peat. This suggests that the presence of alternative electron acceptors does not necessarily completely inhibits methanogenesis under natural conditions due to spatial heterogeneity of these inorganic compounds. However, these Fe and S concentrations do not come from the thermokarst where gas fluxes were measured, and the recent thermokarst site lacks *Sphagnum* cover which was present at the sampled thermokarst, so actual concentrations may differ. Even though, the data provide a useful approximation of the potential availability of inorganic electron acceptors at the site.

Further, main pathways of methane production are acetoclastic methanogenesis, which is based on acetate fermentation, while hydrogenotrophic methanogenesis is based on hydrogen gas as the major reductant (see chapter 1.2). Hydrogenotrophic methanogenesis was found to be the dominant pathway of CH_4 production in northern peatlands, but the acetoclastic pathway might dominate in surface layers in minerotrophic, nutrient-rich peatlands and in young thermokarst stages with recently established vegetation (Heffernan et al., 2022). Hydrogenotrophic methanogenesis was found to be the main pathway in the recent thermokarst at the site (Henrich, 2025). Low inputs of labile organic matter via vascular plants likely explains the limited contribution of acetoclastic methanogenesis in the recent thermokarst, where bare peat without living vegetation predominated (Henrich, 2025).

The water table did not show a significant influence on the emissions from the thermokarst (Table 3). This may be because water table alone does not determine emissions; rather multiple interacting factors govern the response. CH_4 emissions likely peak at an optimal water table position and decline at both higher and lower water table levels (Figure 26), as also found by other studies (Chen et al., 2025; Turetsky et al., 2014). Methane emissions remain close to

zero when the water table lies 0 to 8 cm below the surface, indicating inundated conditions that are related to limited diffusion of CH_4 through standing water (Chen et al., 2025). As soon as the water table reaches a depth of ~ 10 cm below surface level, emissions increase sharply up to 750 mg CH_4 m^{-2} d^{-1}. Fluxes decline again as soon as the water table drops to a lower level of > 10 cm (Figure 26).

Besides hydrological conditions, soil temperature acts as additional control in thermokarst methane emissions since higher temperatures enhance higher emissions under favorable (e.g. anoxic and substrate-rich) conditions (Best et al., 2023; Treat et al., 2015). Methanogenesis typically shows an exponential increase with temperature, as expressed by the Q_{10} relationship, indicating a strong temperature sensitivity of methanogenic activity (Fan et al., 2022). In addition, Lu et al. (2022) showed that increasing temperatures enhanced soil microbial and enzyme activities, changed the community structure and diversity of methanogens and thus accelerated the decomposition of substrates which further promote CH_4 emissions. The fact that temperature had a significant effect on thermokarst methane emissions, but not on methane oxidation in the palsas, may be attributed to the higher temperature sensitivity of methanogens compared to methanotrophic microorganisms (Dunfield et al., 1992). Summarizing, the effects of VWC and soil temperature on the fluxes from the recent thermokarst are indicated to be the main controls and act interactive, suggesting synergistic effects (Olefeldt et al., 2013), while water table only leads to higher emissions under optimal water level conditions.

Nonlinear CH_4 flux slopes were observed intermittently during the 5 min measurement time in the recent thermokarst. These irregular peaks are typical of ebullition, where emission rates are comparable to diffusive fluxes but can induce episodically high rates in waterlogged soils (Baird et al., 2004; Turetsky et al., 2008). This physical gas transport explains the pronounced short-term methane flux variability observed, driven by irregular timing and magnitude of ebullition events.

Compared to reported methane flux rates from other studies summarized in Table 6, the palsa methane flux rates were slightly lower than those reported from organic soil palsas, but fall within the range reported for organic thermokarst ponds. However, two studies (Määttä et al., 2026; Olefeldt et al., 2013) also reported positive fluxes in palsas, which may be attributed to advanced permafrost thaw resulting in increased soil moisture causing methanogenesis. Emmerton et al. (2014) reported both methane uptake and production at the same research site, transitioning to a source of methane after two weeks of the period (during the first two weeks of July: −0.3 mg CH_4 m^{-2} d^{-1}, afterwards until end of growing season: 0.8 mg CH_4 m^{-2} d^{-1}). This temporal shift from sink to source in the polar desert may reflect progressive waterlogging and expansion of anaerobic microsites during the season, whereas the palsas in this thesis permanently supported methane uptake. In contrast to organic soils, some mineral tundra soils with shallow organic layers showed slightly higher methane uptake than organic soils presented in Table 6, likely due to mineral soil properties that promote oxygen supply and thus support

methanotrophic activity. However, other studies from similar environments report methane uptake rates below those, indicating site-specific variations. Soil texture strongly influences water retention and the diffusion of atmospheric CH_4 and O_2 through its control on pore size and connectivity, with sandy soils generally exhibiting higher CH_4 uptake than silt-rich or clayey soils (Murguia-Flores et al., 2018). This suggests that the methane flux rates from arctic mineral tundra soils may vary considerably depending on local soil texture, with sandier patches favoring higher methane oxidation compared to clay- or silt-rich areas. Taken together, these comparisons highlight that methane fluxes in arctic and sub-arctic environments are strongly influenced by soil type, hydrology, and the stage of permafrost thaw, underscoring the importance of site-specific conditions in shaping the balance between methane oxidation and methanogenesis.

Table 6: Average methane fluxes (± SE) reported from permafrost regions. Flux rates marked with * display median values, without * the mean. Not all studies provided SE.

CH_4 flux ***[mg CH4 m^{-2} d^{-1}]***	**Ecosystem type and soil**	**Location**	**Reference**
-2.4 ± 0.2	intact palsa, organic soil	Finnmark, Norway	this study
-2.1 ± 0.2	degraded palsa, organic soil	Finnmark, Norway	this study
80.7 ± 29.2	recent thermokarst, organic soil	Finnmark, Norway	this study
- 3.6 ± 0.3	tundra forest, mineral soil	Kluane Lake, Canada	Juncher Jørgensen et al., 2024
-3.2 ± 1.4	dry tundra, mineral soil	Zackenberg, Greenland	Juncher Jørgensen et al., 2015
-3.1	palsa with Lichen, organic soil	Finnish Lappland	Voigt et al. , 2023
-2.8 ± 0.1	tundra, mineral soil	Disko Bay, Greenland	Juncher Jørgensen et al., 2024
-1.8 ± 1.2	tundra, mineral soil	Kangerlussuaq, Greenland	Juncher Jørgensen et al., 2024
−1.4 ± 0.1	polar desert, mineral soil (with pockets of cryoturbated organic matter)	Ellesmere Island, Canada	Emmerton et al., 2014
-1.4 ± 0.1	tundra, predominantly mineral soil (locally some organic fens)	Disko Island, Greenland	D'Imperio et al., 2017
- 1.2 ± 0.6	moist tundra, predominantly mineral soil with organic layer	Zackenberg, Greenland	Juncher Jørgensen et al., 2015
-0.8	upland tundra with Lichen, organic soil	Trail Valley Creek, Canada	Voigt et al., 2023
0.2 ± 0.1	meadow wetland, predominantly mineral soil with organic layer	Ellesmere Island, Canada	Emmerton et al., 2014
2.2 *	intact palsa, organic soil	Stordalen Mire, Sweden	Määttä et al., 2026

7.0 *	palsa, organic soil	permafrost zones in the Northern Hemisphere	Olefeldt et al., 2013, synthesis paper
31.8 *	partly thawed permafrost, organic soil	Stordalen Mire, Sweden	Määttä et al., 2026
58.1	permafrost wetlands, influenced by variable proportions of organic matter	permafrost zones in the Northern Hemisphere	Chen et al., 2025, synthesis paper
75.8 *	permafrost fen, organic soil	permafrost zones in the Northern Hemisphere	Olefeldt et al., 2013, synthesis paper
76.1 *	fully thawed permafrost, organic soil	Stordalen Mire, Sweden	Määttä et al., 2026

4.2 Influence of thaw depth on methane fluxes in intact and degraded palsas

Hypothesis 2 anticipated that methane oxidation increases as thawing in the active layer proceeded throughout the growing season, since an increasing peat layer would become available for methane oxidation and rising temperatures throughout the season enhancing microbial activity. This hypothesis could not be confirmed.

A possible cause for this non-effect is that increased methane oxidation was cancelled out by increased methane production above the permafrost layer. Deeper peat layers are less infiltrated by oxygen, which promotes methane production under prevailing anaerobic conditions (Kiuru et al., 2022; Voigt et al., 2019). Additionally, as thawing and temperature increase proceed throughout the growing season, previously frozen peat becomes available for microbial decomposition (Liu et al., 2025), and if the expanding thawed layer remains anaerobic, it may further promote methane emissions, thereby counteracting methane uptake. Another reason could be that the effect of thaw depth was superimposed by effects of other soil physical parameters as these might have dominated in affecting methane fluxes (Voigt et al., 2019). Even though soil temperature and VWC did not correlate with palsa methane fluxes, they exhibited higher effect sizes (Table 3), suggesting a stronger influence and thus maybe overriding the thaw depth effect. Besides, under in situ conditions, it is challenging to assess whether permafrost thawing enhances or suppresses methane oxidation or enhances methane production, as thawing simultaneously alters multiple soil properties such as temperature, soil moisture and substrate availability, which in turn subsequently affect methane fluxes (Mauritz et al., 2017). Furthermore, the presence of permafrost itself can impede gas diffusion, limiting oxygen supply (Chamindu Deepagoda & Elberling, 2015) and thus constrain aerobic methane oxidation. Permafrost also influences methanogens and methanotrophs indirectly by regulating hydrological and temperature regimes (Yang et al., 2017). A similar idea is stated by Voigt et al. (2019). In their mesocosm experiment with simulated thawing, even though enhanced CH_4 concentrations were observed at depth after thaw (> 300 ppm vs. ~ 1.9 ppm ambient concentration), a thick oxidative layer avoided CH_4 emissions to the atmosphere in dry and wet mesocosms. This effect can be preserved as long as drainage keeps the layer above thaw sufficiently unsaturated to maintain optimal aerobic conditions for methanotrophs. Concluding, this suggests that the hydrology regime tends to be the main influential driver, with local drainage conditions governing the extent of drying or wetting, while thaw depth acts indirectly to make a thicker peat layer available for microbial activity but also constitutes a diffusion barrier (Chamindu Deepagoda & Elberling, 2015; Voigt et al., 2019).

4.3 CH_4 fluxes in different peat layers under same temperature conditions

The highest methane uptake was expected in the upper ten centimeters of the peat layer, while deeper layers were expected to show decreasing uptake and possibly even slightly methane emissions. In comparison to samples from intact palsas, samples from degraded palsas were anticipated to be below this level.

The most methanotrophic active layer was from 10-20 cm depth in the intact palsa samples. As already explained in chapter 4.1, it is suggested that there is an optimal zone where methane oxidation is highest, which was the 5-10 and 10-15 cm layer in the study by Juncher Jørgensen et al. (2024), roughly matching the findings in this thesis. Methanotrophs seek optimal habitat conditions, which were expected to be found in the 0-10 cm layer, but were seemingly located in 10-20 cm depth in this thesis, where the soil is not too dry compared to the upper layer which may be affected by soil drying, but still provided with enough CH_4 and O_2. This downward shift is thus supposedly attributable to drought stress in the 0-10 cm layer, where surface drying – even if not apparent in this single snapshot of VWC (Figure 12 b) - can induce drought stress from prior drying events that damages methanotrophic cell walls, membranes and enzymes, resulting in reduced microbial activity or entering dormancy (Thomsen et al., 2025). The moisture optimum thus constitutes a state where VWC is high enough to maintain vital functions but also does not impair oxygen supply. Samples originating from deeper layers in intact palsas showed gradually declining methane oxidation rates (lowest rates at maximum depth), attributable to minimal oxygen supply in this layer. As indicated by the results from the PCA and the Pearson correlation, this decline in methane uptake is related to a combination of depth and VWC, as deeper soil layers and higher VWC limit diffusion of oxygen to deeper soil layers and thus inhibit methane oxidation. Depth and VWC were also identified as significant drivers of methane oxidation in the degraded palsa samples, where methane oxidation declined as well with increasing depth. Contrary to intact palsas, the most microbially active layer in the degraded palsa samples was the 0-10 cm zone because optimal conditions might have been found there. This layer exhibited 15 % higher VWC median values than the 0-10 cm zone from intact palsas, which indicates that this zone was not drought-limited and thus able to support peak methanotrophic activity among all degraded palsa layers. The difference in VWC in the 10-20 cm layer compared to intact palsas numbered already 30 % (Figure 12 a), which suggests that deeper layers were already too moist for methanotrophs to perform to their full extent. The consistently higher water content in the degraded palsa samples is thus suggested to be the cause for reduced methane oxidation potential in all layers of soil. Additionally, comparatively higher VWC in the 50-60 and 60-70 cm samples induced minor methane production. These emissions may be related to factors discussed later in this chapter.

The relationship between methane fluxes and VWC detected in the incubation experiment (Figure 27) displays highest methane oxidation at a range of 22 – 50 % VWC. At VWC > 50 %, methane oxidation rates begin to decline; these samples are mostly from degraded palsas

only. As reported in the literature, VWC > 20 % is supposed to reduce CH_4 uptake due to restricted diffusion (Murguia-Flores et al., 2018), and field observations indicate that methane oxidation decreases sharply at VWC > 50 % (Van Den Pol-van Dasselaar et al., 1998). However, these threshold values were derived from mineral soils (sandy grassland soils, temperate forest and boreal sites) and may therefore not be directly transferable to organic peat soils, where gas diffusion and moisture may show different dynamics. Nonetheless, they can serve as approximate reference points for understanding moisture-related limitations on CH_4 uptake in peat environments.

The moisture-dependent pattern also influences methanotrophic community distribution. Methane oxidation is affected by abundance of methanotroph communities, which are suggested to not be equally but abundantly distributed within depths, with increasing abundance in deeper layers, especially in partly thawed bog sites that are frequently below the water table (Singleton et al., 2018). Two main types of methanotrophs have been identified: low-affinity methanotrophs that thrive under high CH_4 concentrations, typically in methane-emitting soils such as thermokarst sites, and high-affinity methanotrophs capable of oxidizing methane even at low concentrations. In palsa sites, low-affinity methanotrophs are predominantly found in the deeper layers (Hanson & Hanson, 1996; Liebner & Wagner, 2007). The occurrence of active methanotrophic communities at depth across Arctic wetland sites (Christiansen et al., 2015; Liebner & Wagner, 2007; Singleton et al., 2018; Tveit et al., 2014), as also observed in this thesis, suggests that these bacteria exhibit a high tolerance to fluctuations in oxygen availability (Blees et al., 2014; Crevecoeur et al., 2017) and are capable of oxidizing methane under microaerobic conditions (Perryman et al., 2020).

Despite this tolerance, methane emissions were detected in deeper layers (Figure 12 a). Detecting methane emissions in the deeper soil layers (Figure 12 a) in an oxic incubation experiment was unusual. Normally, even in anaerobic incubation experiments, emissions are detected only after a long time of incubation, because alternative electron acceptors are depleted prior to start of methanogenesis. For example, in an experiment by Pastukhov & Kaverin (2025), the amount of methane released was extremely low and recorded only in the transitional layer (between active layer and permafrost layer) in depth of 30 – 90 cm after a long time of incubation (~ 500 days), suggesting unfavorable conditions for the development of CH_4-producing microbial communities. Constant emissions from the permafrost layer (70-90 cm) were negligibly low. The low emissions also result from a general limited methane production potential in palsas because the peat is already highly humified and only little amounts of labile carbon are provided by the vegetation. In contrast, dense vegetation in the thermokarst produce large amounts of labile carbon, promoting methanogenesis under anaerobic conditions (Baysinger et al., 2025; Hodgkins et al., 2014).

Methane emissions from deeper originating peat samples in the incubation in this thesis may result from the persistence of anaerobic zones within the peat, as the consistently high

water content prevented the formation of alternative electron acceptors as exposure to oxygen regenerates electron acceptors and suppresses methanogenic activity (Knorr et al., 2009). This suggests that methane was produced during the incubation under anaerobic conditions and potentially also released from previously formed reservoirs. Coexisting oxic and anaerobic microcompartments likely occurred within the peat, with heterogeneously distributed saturated anaerobic regions, mainly concentrated in the inner sections of the samples. Methane emissions in the deeper samples suggest a decreasing potential for atmospheric methane uptake with increasing depth.

However, in the experiment presented in this thesis, number of samples where methane emissions occurred was overall low. In most cases only one or two samples displayed emissions, which had an overriding effect on the mean value towards a positive value. In addition, the incubation was conducted under oxic conditions, which likely led to an underestimation of methane production. However, conditions within the core may still have been relatively close to in-situ environments, allowing for a partial representation of methane flux potential. Despite this limitation, the indicated overall trend of decreasing oxidation with increasing depth was still captured and can be considered robust, as similar patterns have been reported in other studies. Only the mean positive fluxes must be interpreted with caution.

According to the PCA, root biomass was not correlated with methane fluxes (Figure 13). First, roots can enhance methane oxidation in palsas since they extract water from the peat in the transpiration process, leading to reduced water saturated soil and thus enabling an intensified methane oxidation process. Some plant species are equipped with root aerenchymal structures (e.g., *Rubus chamaemorus*, which occurred frequently in the vicinity of the spots where the samples were taken). This trait facilitates oxygen and methane transport into deeper soil layers (Määttä et al., 2026), preventing methanotrophs from substrate and oxygen deprivation (Liebner & Wagner, 2007). On top of that, decomposition of decaying root material releases nutrients that enhance growth of methane oxidizing microbes, promoting a higher microbial activity and therefore a potentially increased methane oxidation rate. Growth conditions and methanotrophic bacteria might also be indirectly connected via roots: optimal temperature and water conditions might support plant and root growth and nutrient cycling, which again impacts methane oxidizing bacteria, as methanotrophs encountered more favorable conditions in the vicinity of the roots. Since several influences are determining the methane flux, effects of other variables (VWC, depth) might be stronger in affecting fluxes and could thus possibly override the effect of roots. Moreover, the roots were disconnected to aboveground biomass, meaning that their full range of natural functions in this artificial setting, such as aerenchymal transportation, were inhibited and could not influence methane cycling to its full extent.

Conversely, the results of the Pearson correlation suggest that root biomass and methane fluxes were negatively correlated. This could originate from the circumstance that VWC and depth are neglected in this calculation and that roots were mostly located in the upper layers

of the peat (as also indicated by Figure 12 c and found by Määttä et al. (2026)), which is due to natural limitations, e.g. peat becomes denser and temperatures decline with increasing depth and despite aerenchymal structures oxygen is limited at a certain depth. At the same time, methane oxidation is coincidentally highest in the upper part of the peat layer and equally declining with depth. This suggests that the actual driver behind this correlation is the depth-gradient. That is further underpinned by a significant correlation between depth ~ fluxes and depth ~ root biomass (not displayed here). Similar results were discovered by Määttä et al. (2026), where no correlation between root biomass and methane flux could be identified. Concluding, this indicates that the actual drivers of methane fluxes across depth are presumably a combination of depth and VWC through influencing substrate supply.

4.4 In-situ assessment of peat layer contributions to methane fluxes in palsas

According to hypothesis 3b, maximum methane oxidation rates were expected in the upper 30 cm of both intact and degraded palsas. Using bottom collars to exclude potential methane input from below 30 cm depth, this hypothesis was not confirmed. By contrast, methane oxidation was higher in deep collars in intact palsas during the season, while degraded palsas showed a similar but delayed pattern from August to early September.

Firstly, little to no methane was likely transported from peat below 30 cm. This is supported by incubations of intact peat segments (see chapter 4.3) where only a few samples from degraded palsas showed net CH_4 production. However, limitations of this experiment regarding methanogenesis must be considered. These positive fluxes were apparently insufficient to affect methane uptake in deep collars, as most methane was likely consumed by methanotrophs under oxic conditions in the overlying peat layers. The lack of significant differences between bottom and deep collar fluxes further supports this conclusion. Secondly, the results were likely restrained by an artefact: VWC in bottom collars exceeded VWC in deep collars. This rise likely resulted from increased precipitation beginning in late July and peaking in August (Figure 15). Rainwater accumulated in bottom-closed collars because drainage holes were applied only late in the season. Precipitation triggered a slightly delayed increase in VWC (Figure 14 c). In both intact and degraded palsas, methane oxidation decreased in bottom collars (Figure 14 a, b), coinciding with peak precipitation in early August.

Previous studies provide additional explanations for these observations. Liu et al. (2025) found that methanogen diversity was significantly lower in the deep anaerobic layers than in the upper oxic and transitional layers, contradicting the assumption that oxygen-limited conditions and anaerobic niches favor methanogens. This may be explained by the proximity of the deep active layer to the permafrost table, where low temperatures and substrate limitations restrict methanogenic activity (Liu et al., 2025). Besides, methanogens are less adapted to low temperatures than methanotrophs (Liebner & Wagner, 2007). Dunfield et al. (1992) also showed that methanogenesis is more temperature sensitive than methanotrophy, with higher temperature optima (25-30°C vs. 20-25 °C) and lower methanogenic activity at low temperatures. Moreover, CH_4 production required higher activation energy (123-271 kJ mol^{-1} vs. 20-80 kJ mol^{-1} for methane consumption) and showed Q_{10} values an order of magnitude greater (5.3-16 vs. 1.4-2.1) (Dunfield et al., 1992). The scarcity of methanogens and substrate and consequently limited methane production may explain why fluxes in deep collars were unaffected by deeper layers.

Further research projects should include repeated measurements in Áidejávri to assess degradation progress and determine how fluxes in degraded palsas have changed compared to intact ones or whether intact palsas have also degraded over time.

4.5 Limitations of the study

Considering the study's limitations, measurements of methane fluxes started mid-July, because thaw depth around 20th of June was insufficient for collar installation. Thus, flux dynamics prior to that time remain unknown, as does the effect of thaw depth on methane fluxes early in the growing season (chapter 4.2). It can be assumed that methane oxidation was lower before mid-July as frozen topsoils show mostly low CH_4 oxidation rates (Voigt et al., 2019).

Furthermore, the fluxes displayed in Figure 12 a (chapter 3.3) derived from the incubation experiment should be interpreted relative to those from other depths rather than as absolute values. Because peat samples were removed from the field and exposed to atmospheric oxygen, only general flux trends can be inferred under ex-situ conditions, where methane uptake may exceed in-situ rates under oxygen- and substrate-limited conditions. In addition, technical constraints prevented maintaining the target water bath temperature of 15 °C, leading to deviations between 13.9 °C and 16.9 °C in the peat samples. These deviations likely introduced additional variability in microbial activity and CH_4 fluxes. Aside from that, the methane oxidation potential may not be fully captured due to heterogeneous spatial distribution of CH_4-oxidizing bacteria within the soil profile, as localized microsites of activity may be missed in core-based incubations. This spatial variability can contribute to differences between chamber-based and sub-surface flux measurements (Juncher Jørgensen et al., 2024), although the use of six replicates likely reduced its overall influence. Despite these limitations, a clear depth-related gradient emerged, driven mainly by decreasing oxygen availability and microbial adaptation to in-situ conditions. Finally, as the incubation experiment was conducted only once at the end of the growing season, the behavior of peat samples before or after that time remains unknown.

Investigating the influence of roots on methane oxidation by measuring root weight might not have been most useful (chapter 4.3). Other root functional traits could have given more insight into relationships. For instance, additionally to the weight, Määttä et al. (2026) determined the root surface area, diameter, tissue density and specific root length and as well sampled rhizomes. They found that root tissue density was strongly and negatively correlated with CH_4 fluxes and increasing herbaceous rhizome surface area was associated with higher CH_4 fluxes, while biomass did not show a relationship to methane fluxes.

As outlined in chapter 4.4, the experimental setup did not allow the intended demonstration because of artefactual influences. Drainage holes in the bottom collars should have been installed earlier to prevent water accumulation and the resulting artificial increase in VWC. Nevertheless, the results reaffirm that high water content - as a proxy for limited oxygen and substrate supply - strongly reduces methane oxidation, as discussed in earlier chapters. To achieve more precise and reliable results, an improved experimental design is recommended for future studies. Whether methane is produced in deeper peat layers can be identified by analyzing methane concentrations in pore water samples from consecutive deeper peat layers, ideally over the growing season. This can be done by e.g. using stainless steel or PVC capillaries equipped

with membranes on one end to prevent clogging by peat. Each capillary should terminate at a distinct depth. A valve at the surface end of each capillary allows pore water extraction. Afterwards, concentration in the pore water can then be determined by incubation in a gas chromatograph. This kind of experiment setup has been used in peatlands in several former research projects (Hornibrook et al., 2009; Strack & Waddington, 2008). The results of methane concentration at each depth can then be used to reconstruct concentration gradients and hence determine whether methane is likely to be emitted to higher, well aerated horizons, where it leads to reduced methane oxidation rates.

5 Conclusion and outlook

Both, intact and degraded palsas functioned as net sink for atmospheric methane, with no differences, suggesting that disturbance in degraded palsas had little or no impact on sink strength. In the palsas, neither soil temperature nor VWC affected fluxes, suggesting that short-term temperature variations were too small to create an impact. Regarding VWC, the findings point to a decoupling between topsoil conditions and CH_4 fluxes, as CH_4 oxidation in deeper layers likely determines surface flux rates. Likewise, thaw depth did not affect palsa methane fluxes, possibly because enhanced methane oxidation rates were offset by increased methane production above the permafrost layer. In recently formed thermokarst areas, persistently moist conditions favored methanogenesis. Here, emissions were significantly affected by VWC and soil temperature, while water level had no clear influence because methane emissions peaked at an intermediate water level and declined at both higher and lower positions. Ex-situ incubations assessing the effect of depth on methane oxidation revealed the highest methanotrophic activity in the 10–20 cm layer of intact palsas and in the 0–10 cm layer of degraded palsas. The observed vertical pattern in methane oxidation was primarily driven by VWC and depth together, as both factors affected oxygen availability by diffusion. Additional research is required to study in which peat depth and how much methane is produced or oxidized over the growing season.

As thawing of permafrost peatlands continues under climate change, shifts in hydrology, soil thermal regimes, weather conditions and other interacting environmental factors are likely to further alter the balance between methanogenesis and methane oxidation. Further degradation of already degraded palsas can lead to thermokarst formation and thus act as a net methane source, mainly depending on moisture conditions of the peat, while their hydrological and vegetation heterogeneity may result in substantial variability in methane emissions. To evaluate the future climate impact of these ecosystems, further research will be essential to deepen the understanding of methane fluxes from thawing permafrost peatlands.

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8 Appendix

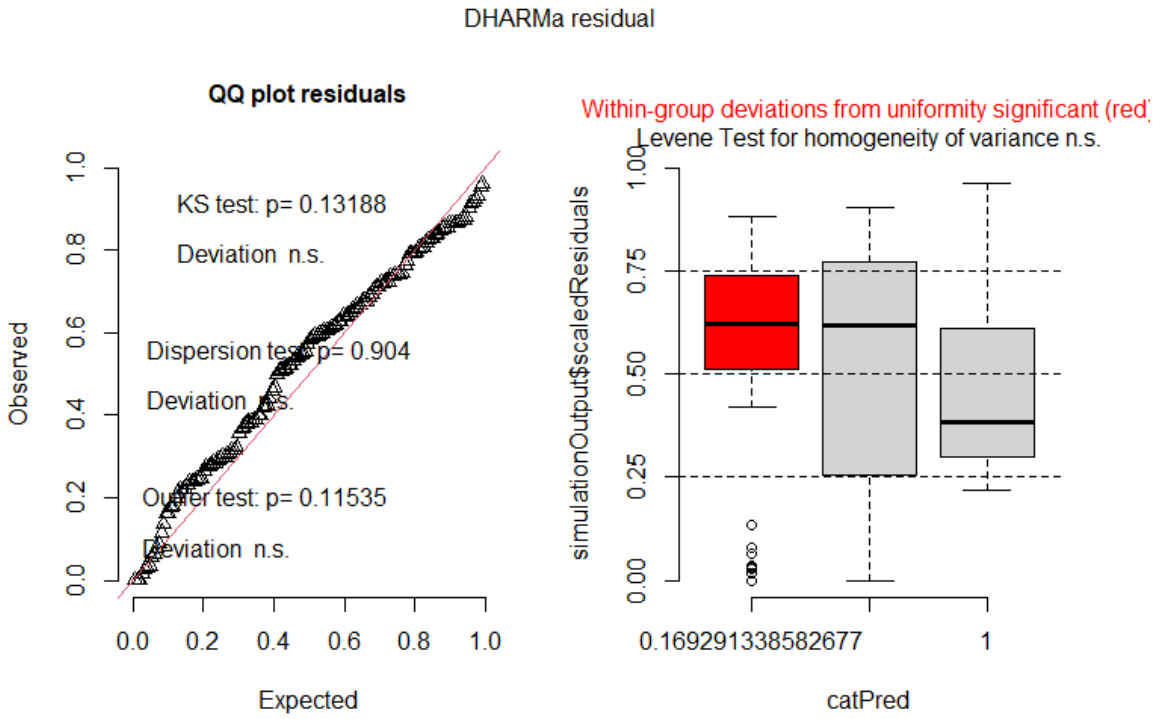


Figure 16: DHARMa residual model plot for hypothesis 1 to test for significant differences between intact and degraded palsas and recent thermokarst structures. The methane flux and treatment-type (intact / degraded palsa & thermokarst) were set as fixed effects, while Collar-ID and Date act as random effects.

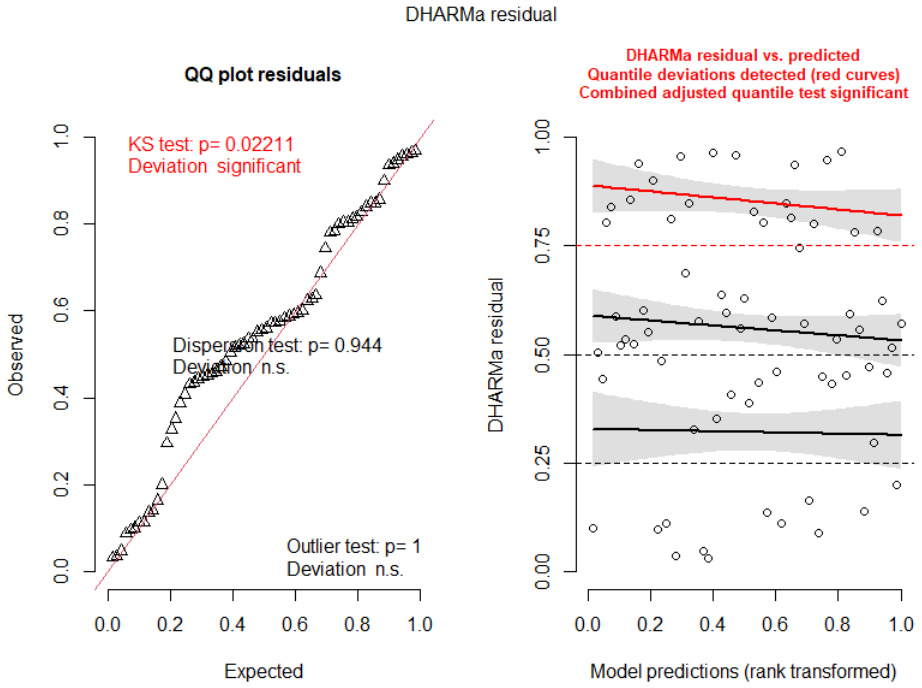


Figure 17: DHARMa residual model plot for hypothesis 1 to test the influence of soil physical parameters on fluxes in intact and degraded palsas. The flux, treatment-type, thaw depth, soil temperature and VWC were set as fixed effects, while Collar-ID and Date were set as random effects.

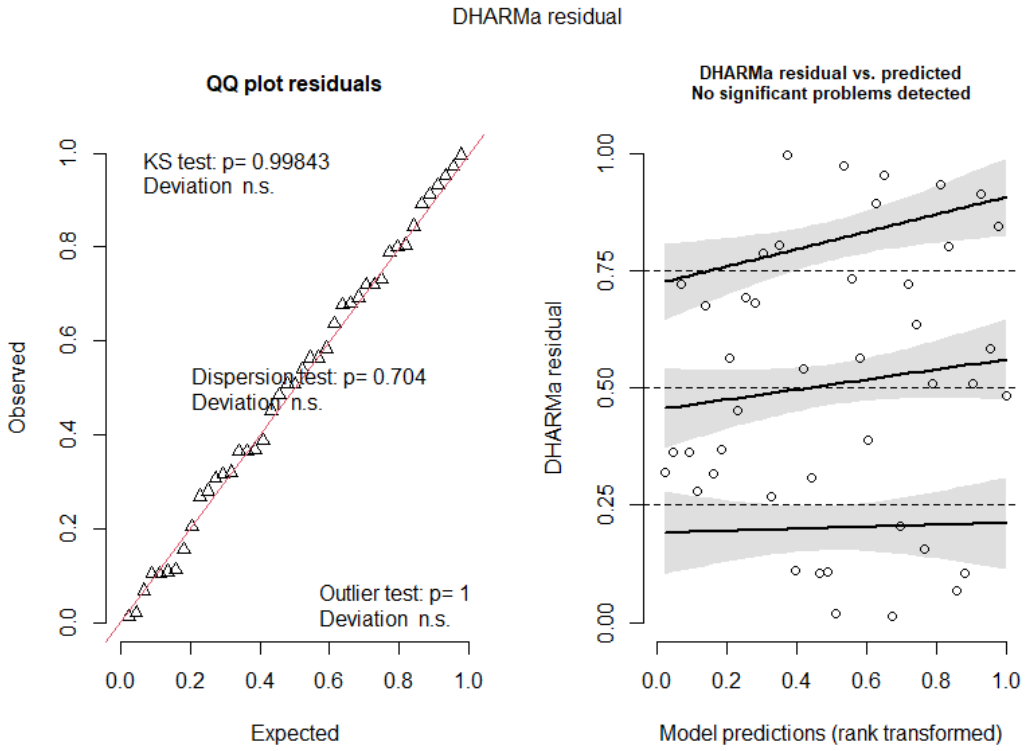


Figure 18: DHARMa residual model plot for Hypothesis 1 to test the influence of soil physical parameters on fluxes from recent thermokarst. The flux, VWC, soil temperature and water level were set as fixed effects, while Collar-ID and Date were selected as random effects.

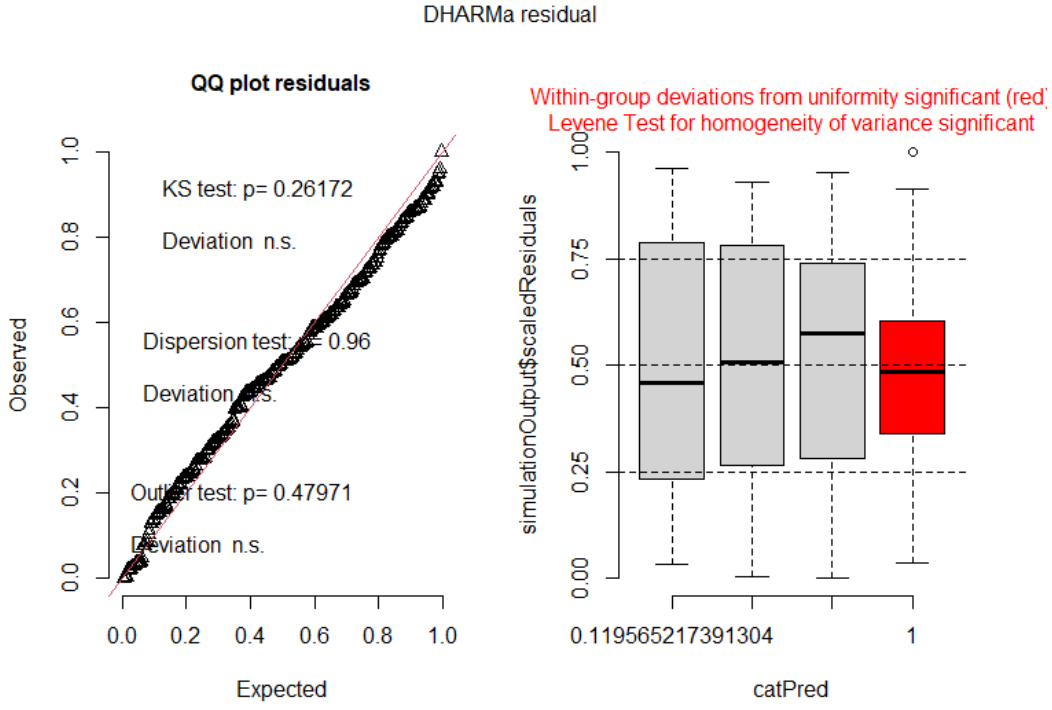


Figure 19: DHARMa residual model plots for hypothesis 3b to test for significant differences between bottom and deep collars from intact and degraded palsas. The flux, treatment-type, soil temperature and VWC were set as fixed effects, while Collar-ID and Date were set as random effects.

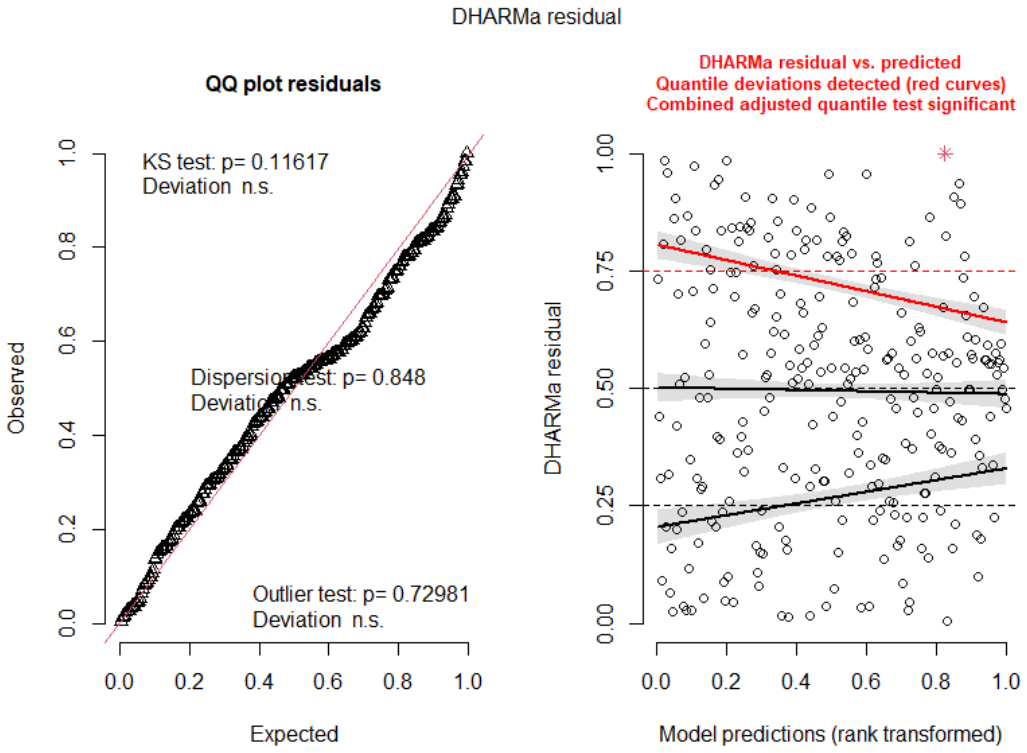


Figure 20: DHARMa residual model plots for hypothesis 3b to test the influence of soil physical parameters on bottom and deep collars from intact and degraded palsas. The flux, treatment-type, volumetric water content (as interaction with treatment-type) and soil temperature were used as fixed effects, while Collar-ID and Date were set as random effects.

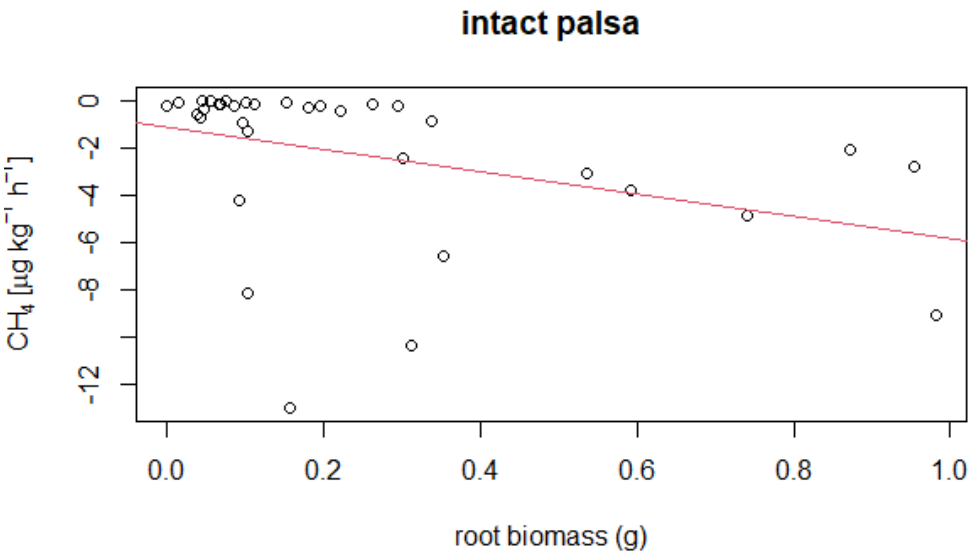


Figure 21: significant correlation between root biomass and fluxes from intact palsa peat samples.

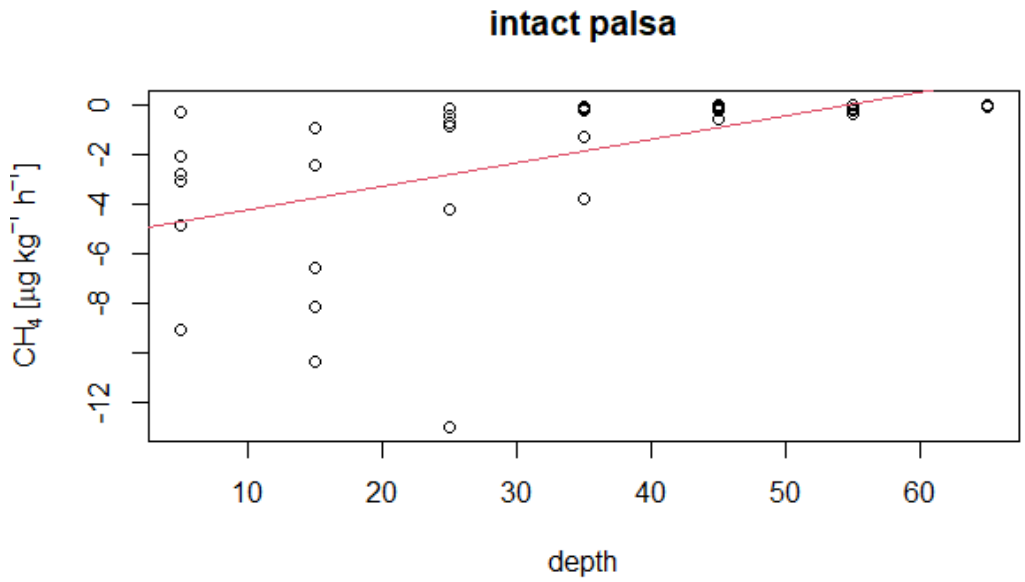


Figure 22: significant correlation between depth and fluxes from intact palsa peat samples.

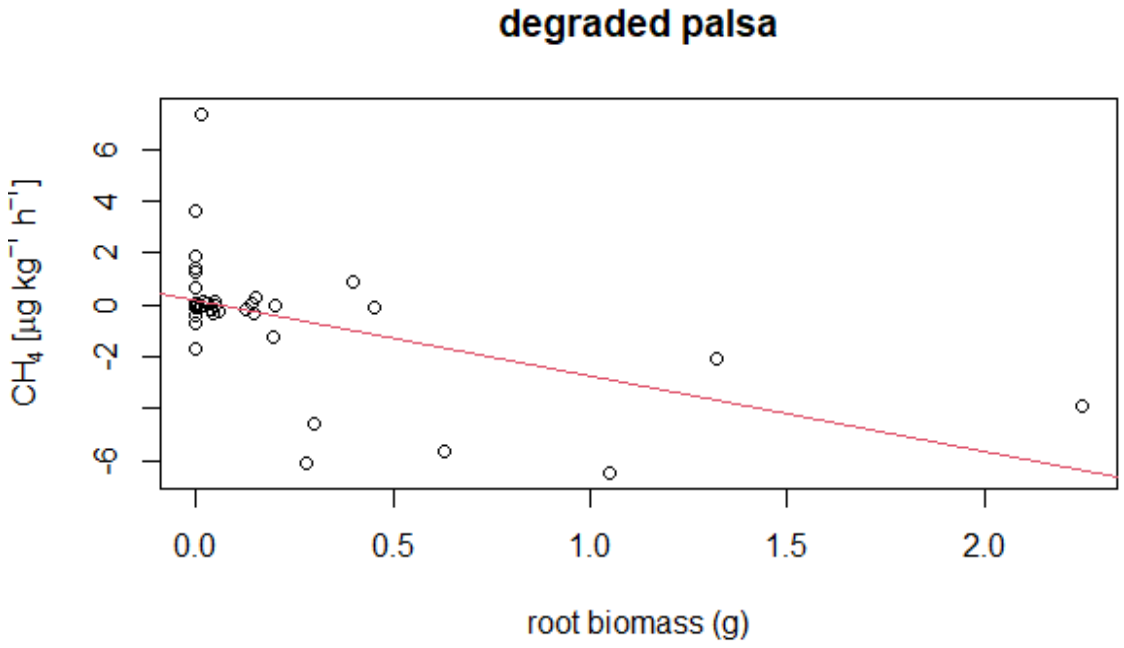


Figure 23: significant correlation between root biomass and fluxes from degraded palsa peat samples.

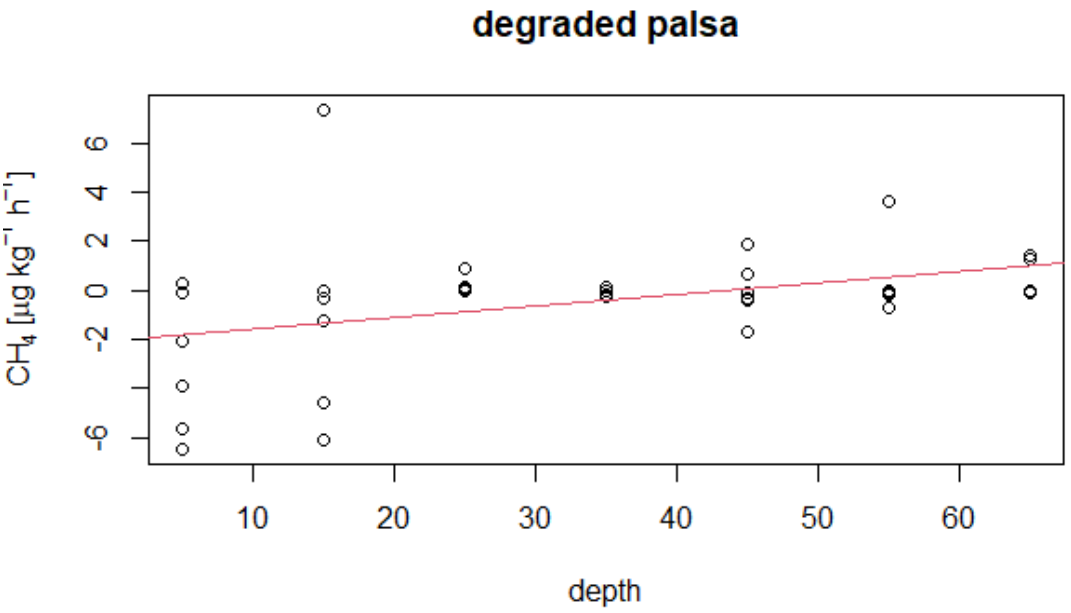


Figure 24: significant correlation between depth and methane fluxes from degraded palsa peat samples.

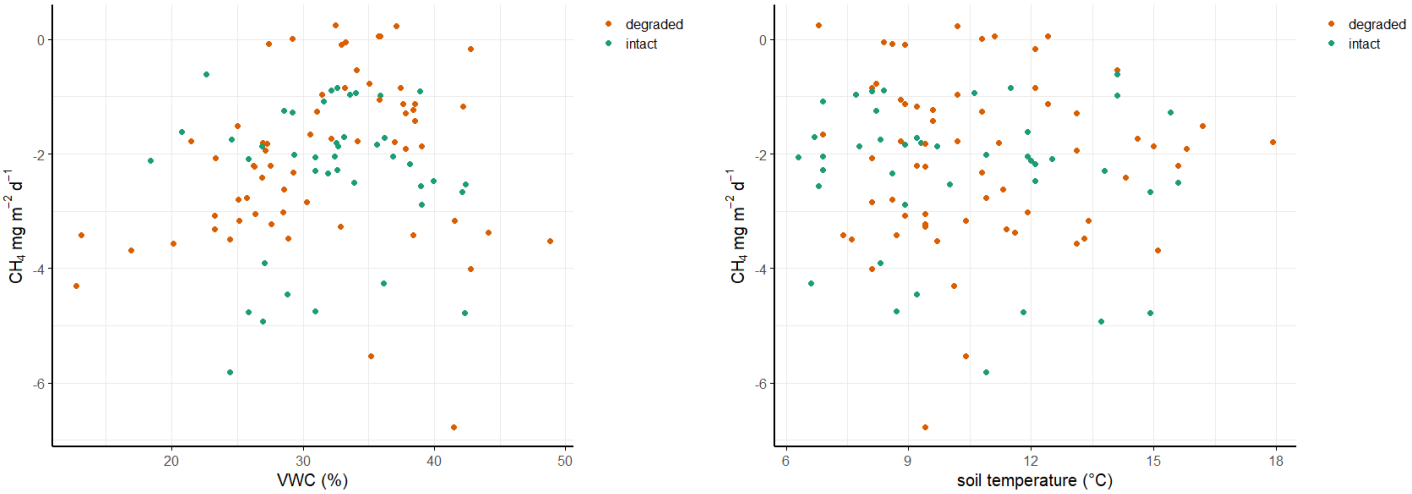


Figure 25: Correlation between VWC (%) and in-situ methane fluxes in palsas (left) and relationship between soil temperature (°C) and in-situ methane flux in palsas (right).

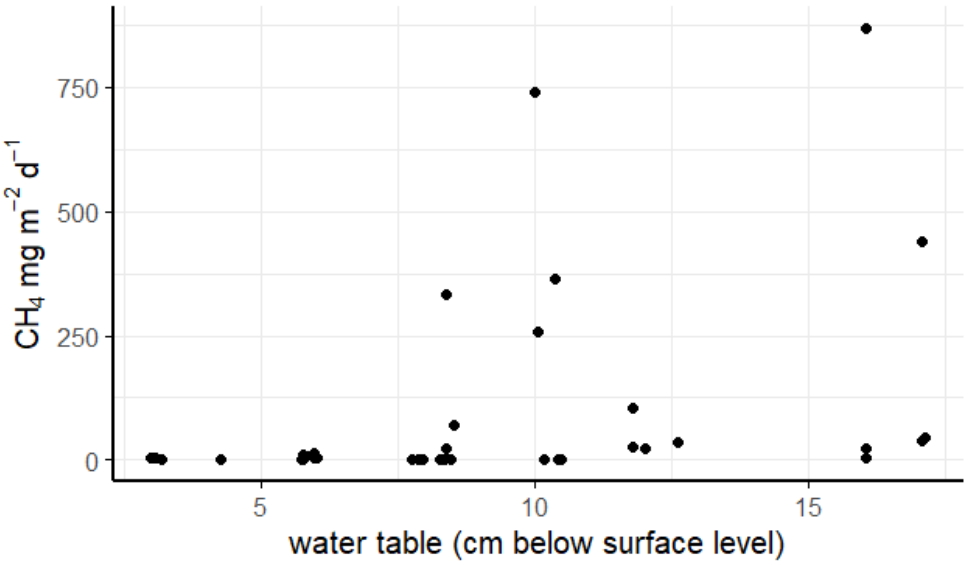


Figure 26: Correlation between recent thermokarst CH_4 flux (mg CH_4 m^2 d^{-1}) and water table depth (cm) to display the relationship between water table depth and growing season methane flux in the recent thermokarst.

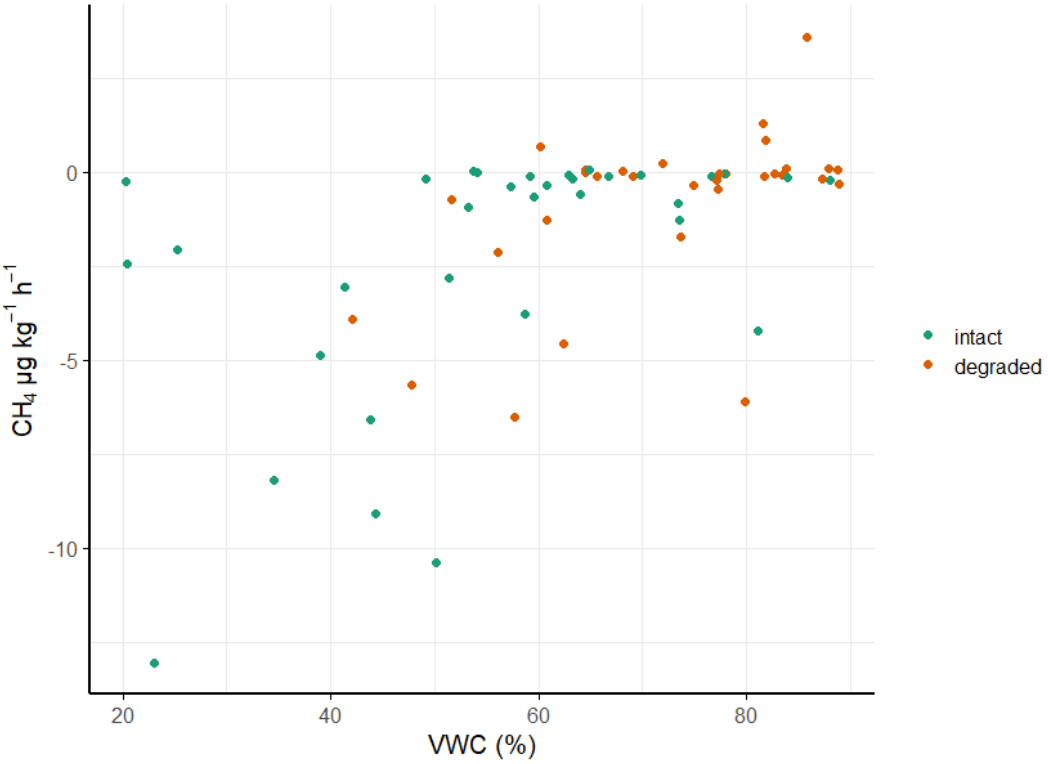


Figure 27: relationship between VWC and methane flux as discovered in the incubation experiment.